CNS Medchem

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A. CNS Depressant Drugs

1. Sedatives and hypnotics
2. Anticonvulsant drugs
3. Major tranquilizers (neuroleptics)
4. Minor tranquilizers (anxiolytics)
5. Central skeletal muscle relaxants

- Drugs affecting the CNS provide
  - Anesthesia, treats psychiatric disorders, relieve anxiety, provide sleep or sedation, prevent epileptic seizure, and suppress movement
  - Lipid soluble non-ionized molecules pass readily into the CNS (BBB)

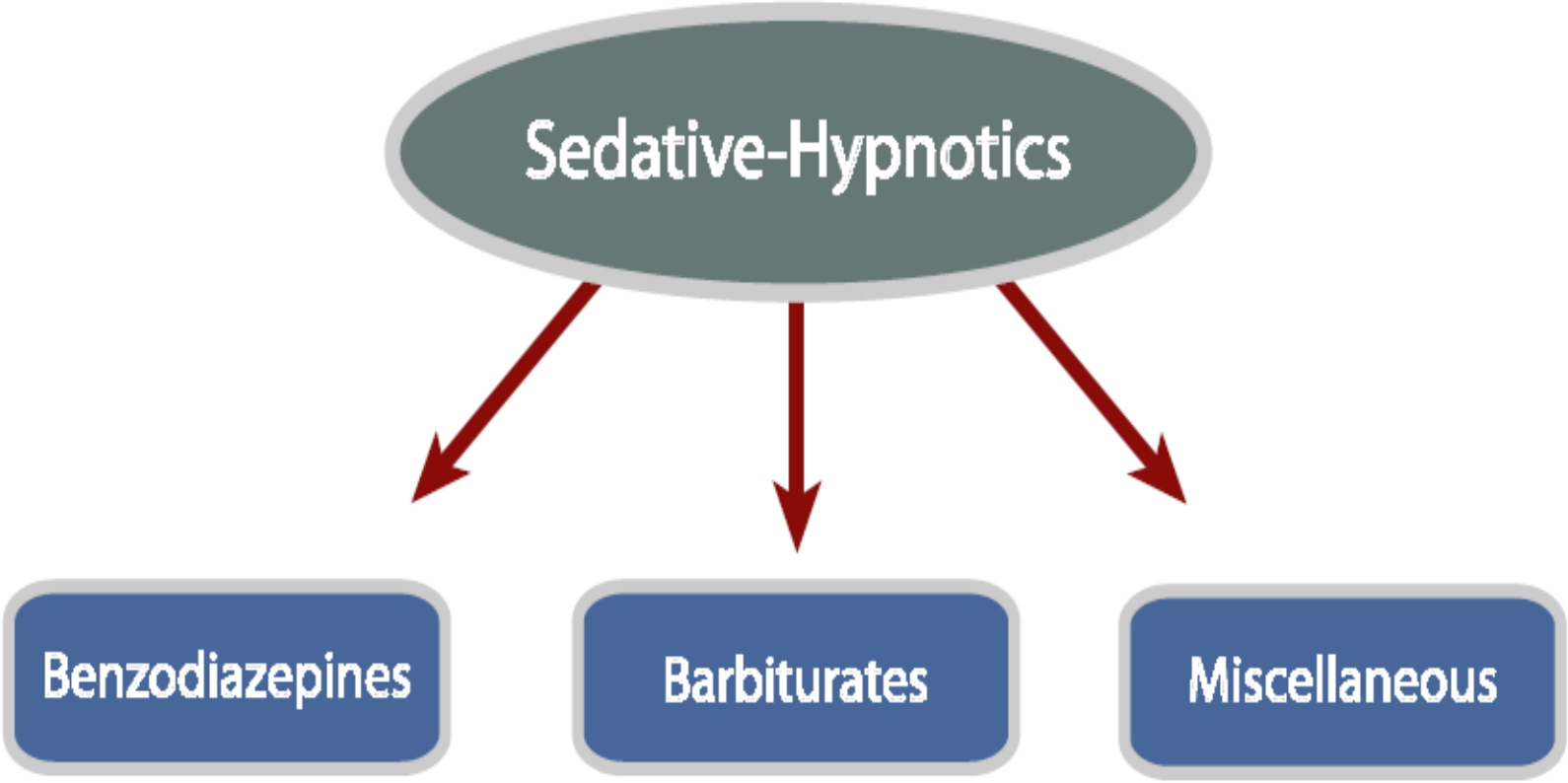
1. Sedative and hypnotics

- Sedatives are drugs which exert a quietening effect accompanied by relaxation and rest but do not necessarily induce sleep. Hypnotics are drugs which induce sleep. Both hypnotic and sedative properties usually reside in the same drug ; a large dose of a drug may act as a hypnotic, whereas a small dose of the same drug would act as a sedative.
- Also employed as anticonvulsant, muscle relaxant and anti-anxiety

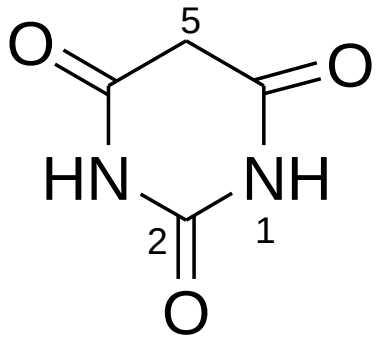
- An ideal hypnotic should
  - Cause transient decrease in the level of consciousness for the purpose of sleep without lingering effect
  - Have no potential for decreasing or arresting respiration (even at high dose)
  - Produce no
    - Abuse
    - Addiction
    - Tolerance or
    - Dependence

Classification of Sedative-Hypnotics

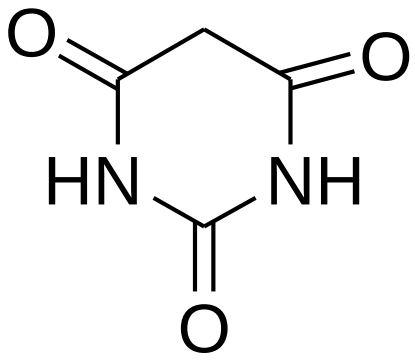
- The sedative-hypnotics may be divided into three major groups:
  - Barbiturates
  - Benzodiazepines
  - Miscellaneous:- drugs that do not fall into either of the first two groups

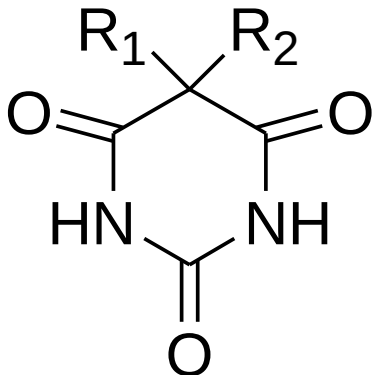

Sedative-Hypnotics
Benzodiazepines
Barbiturates
Miscellaneous

i. Barbiturate



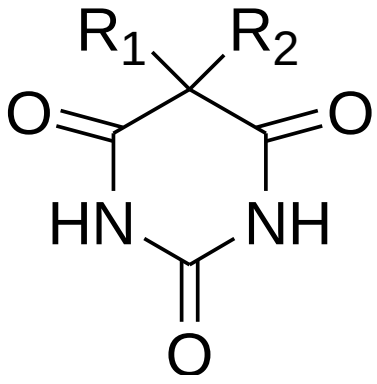
- Barbituric acid was synthesized in 1864 (at the day of St. Barbara) by a German chemist, Adolph von Baeyer from a reaction of urea with malonic acid, a chemical found in apples.
- But was not used as a drug



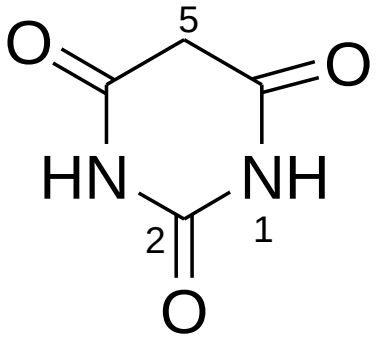


Drugs	R_1	R_2	Branch at N1	Duration
Phenobarbital	$-CH_2CH_3$	$-C_6H_5$	-	Long
Mephobarbital	$-CH_2CH_3$	$-C_6H_5$	CH_3	Long
Barbital	$-CH_2CH_3$	$-CH_2CH_3$	-	Long
Amobarbital	$-CH_2CH_3$	$-CH_2CH_2CH(CH_3)_2$	-	Intermediate
Allobarbital	$-CH_2CH=CH_2$	$-CH_2CH=CH_2$	-	Intermediate
Butabarbital	$-CH_2CH_3$	$-CH(CH_3)CH_2CH_3$	-	Intermediate
Pentobarbital	$-CH_2CH_3$	$-CH(CH_3)CH_2CH_2CH_3$	-	Short
Secobarbital	$-CH_2CH=CH_2$	$-CH(CH_3)CH_2CH_2CH_3$	-	Short
Cyclobarbital	$-CH_2CH_3$		-	Short
Thiopental	$-CH_2CH_3$	$-CH(CH_3)CH_2CH_2CH_3$	-	Short

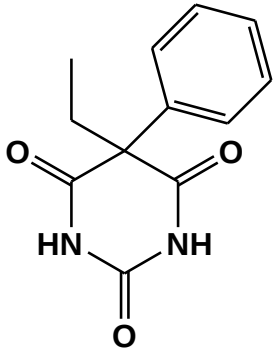
SAR



- Hypnotic activity increases with lipid solubility until the total number of carbon atoms at both C-5 substituents is between 6 and 10.
  - Further increase in the sum of the number of carbon atoms decreases hypnotic activity despite increased lipophilicity, indicating that lipophilicity must remains within certain limits.
- Within the same series, the branched chain isomer generally has greater lipid solubility, hypnotic activity, and shorter duration of action than the straight chain isomer.

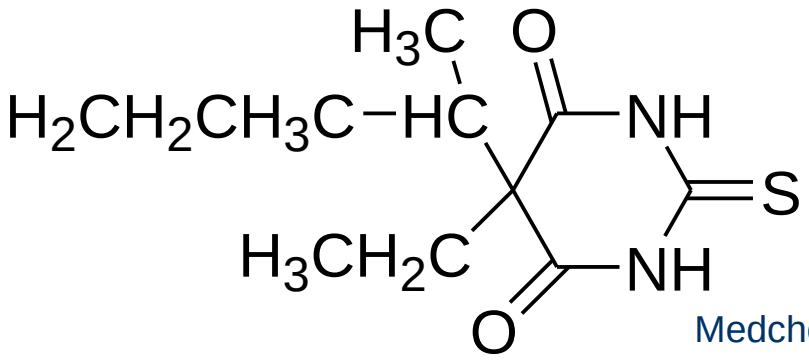


- Within the same series the unsaturated allyl, alkenyl, and alkynyl derivatives are more hypnotic than the saturated analogs with the same number of carbon atoms.
- Conversion of a 5, 5-disubstituted barbituric acid to a 1, 5, 5-trisubstituted analog does not result in a significant change in hypnotic activity.



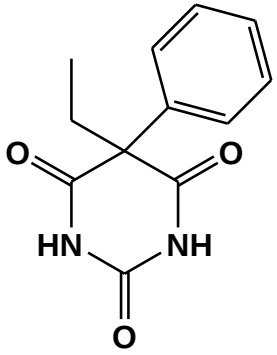
Phenobarbital

- Introduction of polar substituents (OH, NH,, COOH, CO, RNH, SO, H) into an aromatic moiety at the 5-position decreases lipid solubility and potency.
- Replacement of the oxygen at C-2 by a sulfur atom results in thiobarbiturates with faster onset but shorter duration of hypnotic activity.

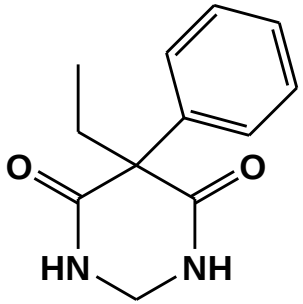


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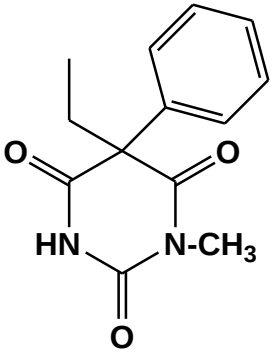
- Aromatic substituents at C-5 produces anticonvulsant activity
  - Barbiturates used as antiseizure drugs



Phenobarbital

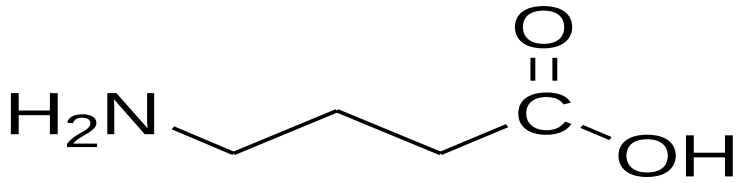


Primidone



Mephobarbital

- Barbiturates bind to GABA receptors on the allosteric site.

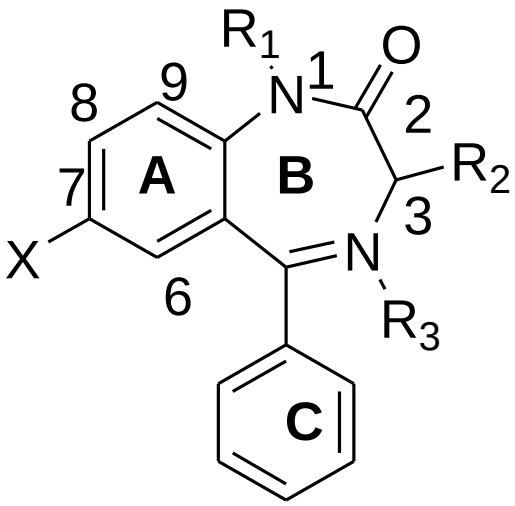


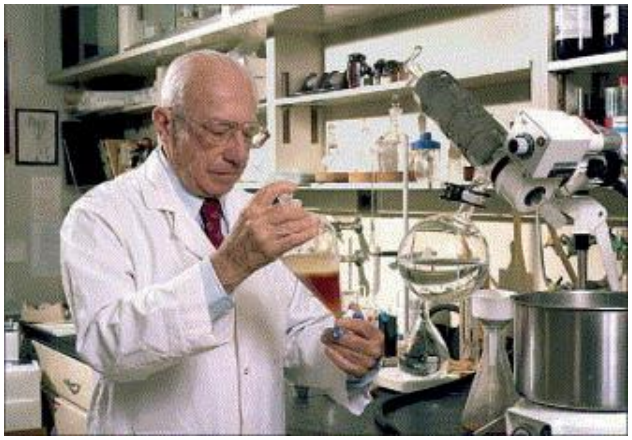
GABA-γ amino butyric acid

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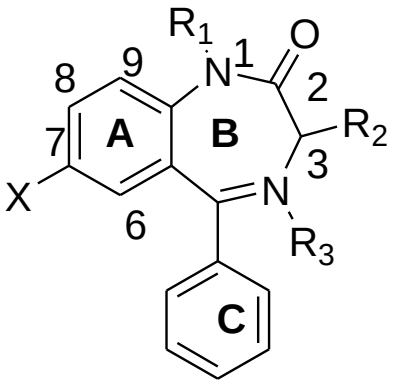
ii. Benzodiazepines

- Benzodiazepine refers to portion of the structure of benzene fused to seven membrane diazepine ring

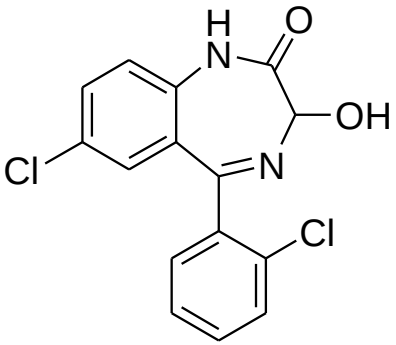




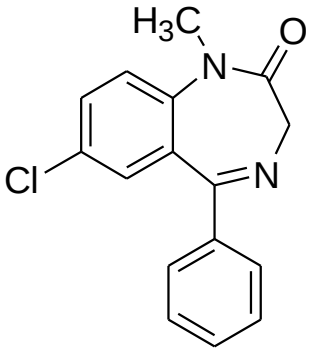
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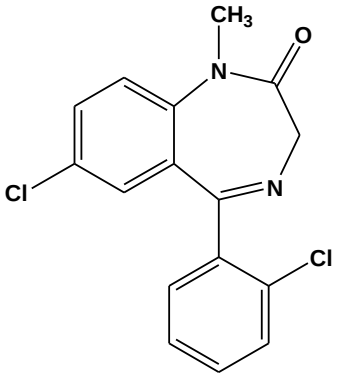
- Substituents at position marked R1, R2, R3 and X result in different drugs



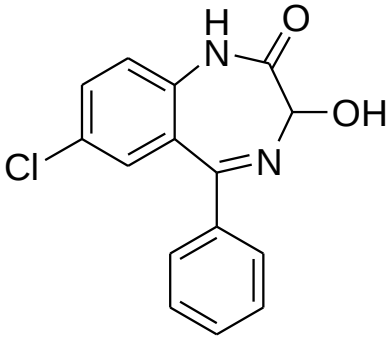
Lorazepam



Diazepam



Chlorozepam



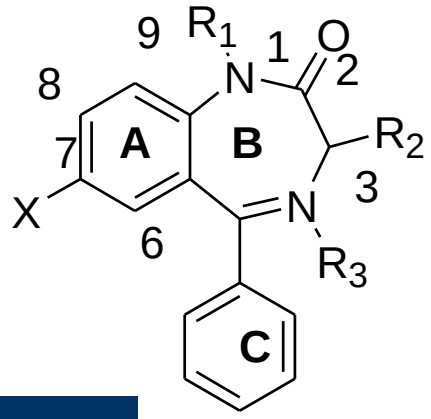
Oxazepam

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- The pharmacological effects of benzodiazepines are outlined as follows:
  - They produce sedation, hypnosis, decreased anxiety, muscle relaxation, and anticonvulsant activity.
  - Benzodiazepines are not general neuronal depressants as are barbiturates.
  - Benzodiazepines have similar pharmacologic profiles, but the drugs differ in selectivity. The clinical usefulness of benzodiazepine drugs varies accordingly.
  - Benzodiazepines enhance GABAergic transmission in all CNS structures.

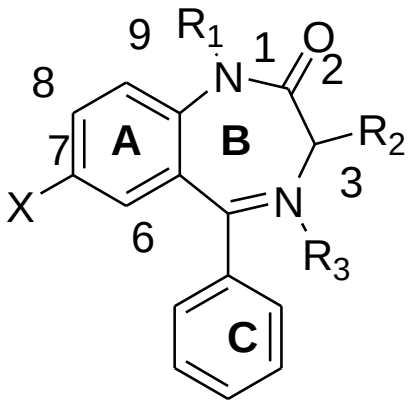
- Benzodiazepines show fewer tendencies to tolerance and dependency than other older sedative-hypnotic drugs, especially barbiturates.
- Also, benzodiazepines produce less abuse potential.
- Benzodiazepines are safer in overdose especially compared with barbiturates.
- Benzodiazepines produce fewer drug interactions because they do not induce hepatic microsomal enzymes

SAR of benzodiazepines



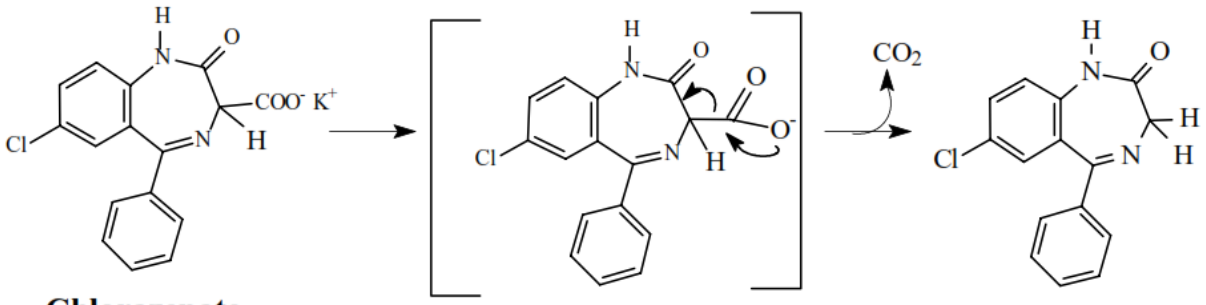
- Rings A, B and C are required for BDZ-receptor binding activity
- Saturation of the 4, 5 double bond or shift to 3, 4-position decreases activity
- Position 6, 8 and 9 should not be substituted
- The 2-carbonyl function is optimal for activity as is the nitrogen atom at position 1. The N substitution should be small , H, Methyl, Ethyl
- An electron withdrawing group at R7 (X= Cl < Br < F < CN < CF_3 < NO_2) is required for optimal receptor affinity

  -The more electrons attracting, the higher the activity
- Substitution on other positions of this ring may decrease activity
- Differences in duration are accounted by structural differences involving primary N-1 and C-3 substitution



- A phenyl group at the C5 promotes activity
  - Ortho or di-ortho substitution with halogen atom increases activity. Para Substituents decrease activity
- Substituents at positions 1, 3 and to a lesser extent 7 influence pharmacokinetics
- Hydroxyl group at C3 is important pharmacokinetically
  - Compounds with alkyl group have longer half-life and less potency
  - Compounds with ester can used as prodrugs

- The 3-carboxylate benzodiazepines are unique in that they contain a 3-COO functionality which allows for water solubility. These drugs, of which **chlorazepate** is the prototype, function as water soluble prodrugs.
- When administered (orally) they are readily protonated in the upper GI tract and spontaneously decarboxylate (loss of CO_2) as shown below to yield an active benzodiazepine which is absorbed from the gut



iii. Other sedative hypnotics (miscellaneous)

A. ALCOHOLS

- Actions: Acts at the $GABA_A$ receptor to potentiate the action of GABA.
- Also inhibits NMDA (N-Methyl-D-Asparate) glutamate receptors.
- These actions result in an array of actions including CNS depression.

ALCOHOLS…

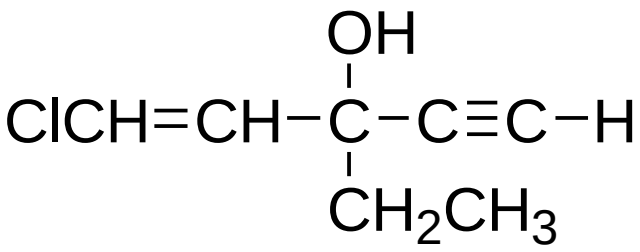
SAR for hypnotic activity of alcohols

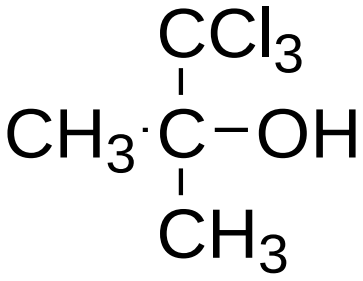
- Activity increases as the chain length increase with maximum at C8
- Branching the chain increases activity

  Tertiary > Secondary > Primary
- Replacing hydrogen with a halogen increases activity

Ethchlorvynol and Chlorobutanol

- A tertiary, lipophilic alcohol

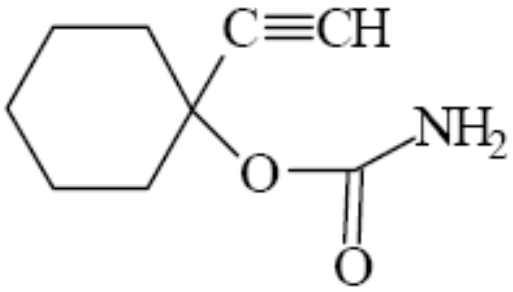




B. Carbamates

Ethinamate

- Carbamate analogue of acetylenic alcohol

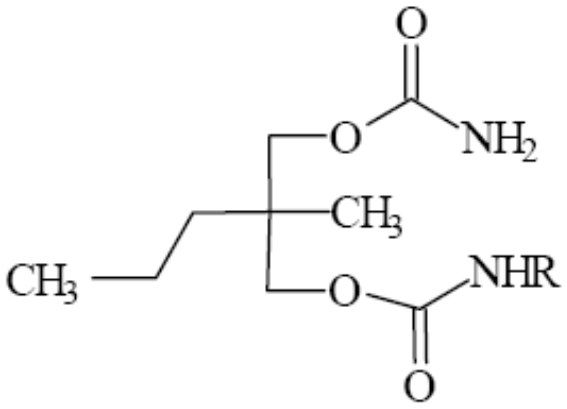


- It is a short-acting hypnotic to which tolerance rapidly develops; it is indicated only for periods of 7 days or less.

Carbamates…

Meprobamate and carisoprodol

- A bis-carbamate–1,3-propanediol derivative

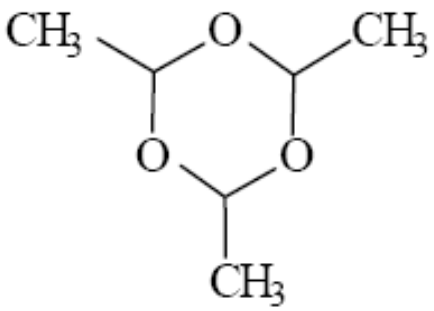


R = H: Meprobamate
R = $CH(CH_3)_2$: Carisoprodol

C. Aldehydes

Paraldehyde

- A trimer of acetaldehyde

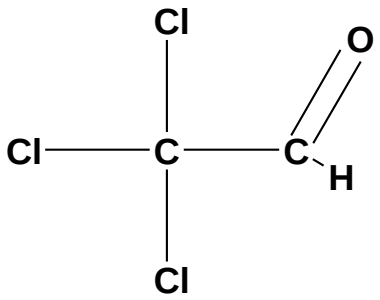


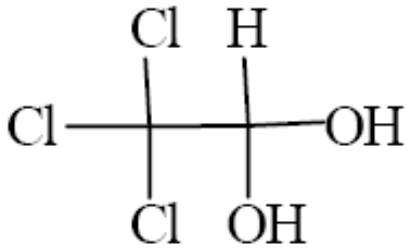
- It produces nonspecific, reversible depression of the CNS and has been used as a rapid-acting hypnotic

Aldehydes…

Chloral hydrate

- It is the hydrate of trichloroacetaldehyde (TCA) and thus is a relatively lipophilic compound.
- In *vivo* this compound is readily hydrolyzed by a chemical mechanism to TCE.
- TCA is reduced by ADH to **trichloroethanol** (TCE), the principal active metabolite.



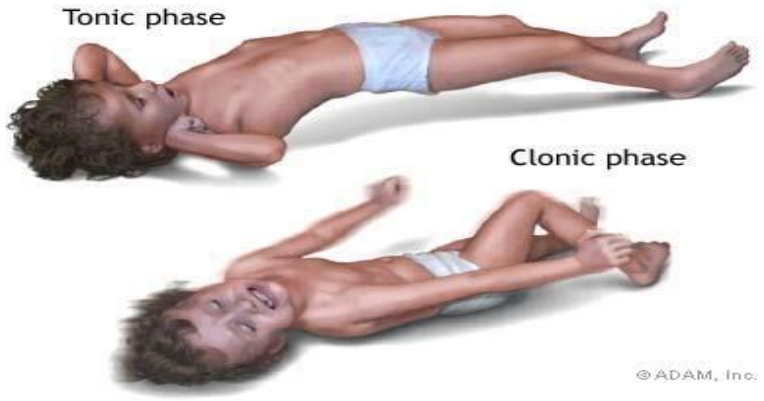


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2. Anticonvulsants (anti- epileptics)



- CNS disorder characterized by recurrent abnormal discharge of CNS neurons
- Characterized by abnormal and excessive EEG (electroencephalography) and discharge
- May be limited to a focus or encompass wide area
- Causes disturbance of consciousness and hyperactivity of autonomic nerves



Anticonvulsants…

- There are four types of epilepsy. Certain signs and symptoms characterize each type

(a) Grand Mal

- *Grand Mal is the most common type of epilepsy. In this type of epilepsy,*the person often experiences an aura (this can consist of certain sounds, fear discomfort) immediately before a seizure. Then the patient loss consciousness and has tonic-clonic convulsions. The seizures generally last from 2 to 5 minutes.

(*b*) ***Petit Mal***

- This type of epilepsy is most frequently found in children. Brief periods of blank spells or loss of speech characterizes petit mal. During the seizures, which usually last from 1 to 30 seconds, the person stops what he is doing and after the seizure resumes what he was doing before the seizure. Many persons are not aware that they have had a seizure.

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(*c*) ***Jacksonian (Focal)***

- *This type of epilepsy is rare. It is usually associated with lesion* of a certain part of the brain (cerebral cortex). Jacksonian epilepsy is characterized by focal or local clonic type convulsions of localized muscle groups (for example, thumb, big toe,and so forth). The seizures normally last from 1 to 2 minutes.

(*d*) ***Psychomotor***

- Psychomotor epilepsy is rare. It is characterized by periods of abnormal types of behavior (for example, extensive chewing or swallowing). Psychomotor seizures occur most often in children 3 years of age through adolescence. The individual may experience an aura with perceptual alterations, such as hallucinations or a strong sense of fear. The localized seizures may advance to generalized convulsions with resultant loss of consciousness.

Anticonvulsants

- Anticonvulsant acts selectively as depressant of convulsant activity in the brain
- The only effective way to control epilepsy is the use of drugs
- They all have common structure features
- Possible MOAs
  - Augment the action of GABA
  - Inhibit calcium or sodium channels
  - Block NMDA (N-Methyl-D-Asparate) receptors

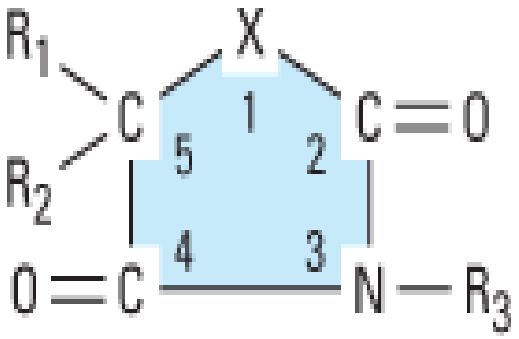
Anticonvulsants…

- **Classification**
  - Cyclic ureides and imides
  - Urea derivatives
  - Miscellaneous

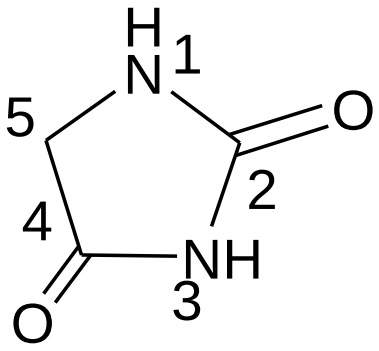
a) Cyclic ureides and imides

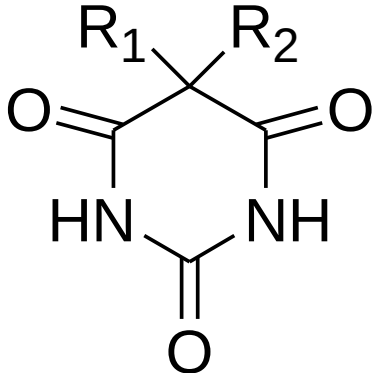
- If both are lower alkyl, the tendency is to be active against petit mal
- If one is an aryl group, activity tends to be directed to grand mal epilepsy
- The strength of activity depends on the substituents

Class	X
Barbiturate	NH
Hydantions	NH
Oxazolidindiones	O
Succinimides	CH_2

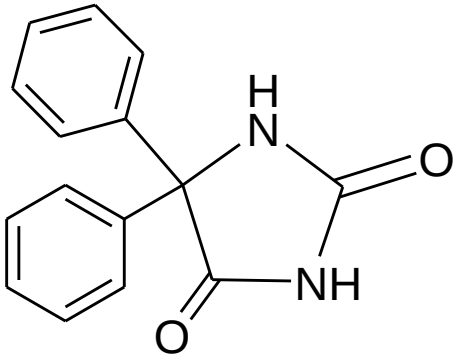


Hydantoins…

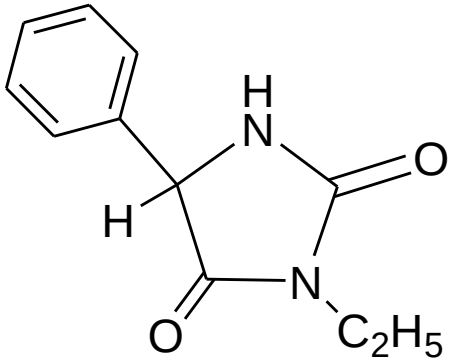




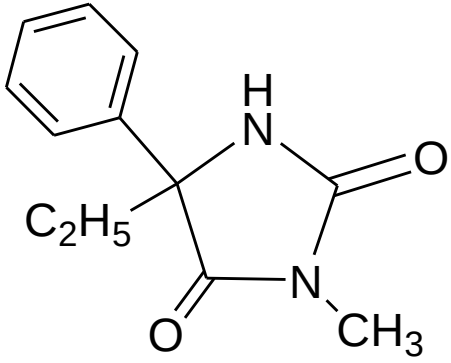
Barbiturate



Phenytoin



Ethotoin



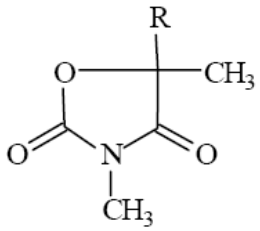
Mephenytoin

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Hydantoins…

- Phenytoin and the other hydantoins are “ringcontracted” analogues of the barbiturates (one carbonyl removed).
- These compounds have physicochemical properties similar to the barbiturates, including acidity when R = H as in phenytoin (an acidic imide with pKa 8.3).
- They are appeared to be act by inhibiting voltage gated sodium channels
- phenytoin is also a weak antiarrhythmic

Oxazolidinediones



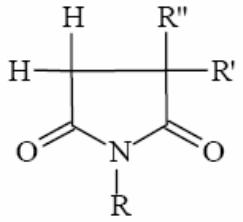
Oxazolidinediones

Trimethadione: R = CH_3

Paramethadione: R = CH_2CH_3

- Trimethadione and the other oxazolidinediones are “ring-contracted” analogues of the barbiturates (one carbonyl removed) or can be viewed as isosteres of the hydantoins (one ring N replaced with O). These compounds have physicochemical properties similar to the barbiturates, except they are not acidic (no imide).

Succinimides



Succinimides

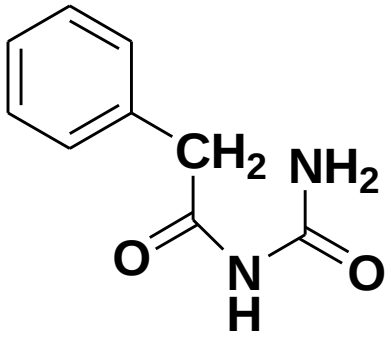
Ethosuximide: R = H, R' = CH_3, R'' = CH_2CH_3

Methsuximide: R = CH_3, R' = CH_3, R'' = 

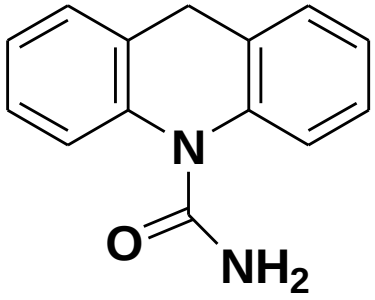
Phensuximide: R = CH_3, R' = H, R'' = 

- Also an isoster of barbiturates
- These compounds have physicochemical properties similar to the barbiturates, except only ethosuximide has an imide N-H and is acidic.
- Active against absence siezure
- The antiepileptic (petit mal) actions of the succinimdes may be related to synaptic inhibition brought about by the GABA mediated chloride conductants

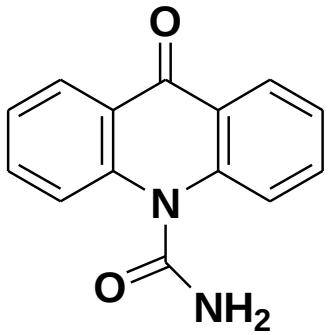
b) Ureas



Phenacemide

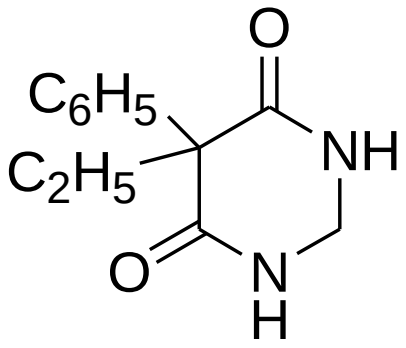


Carbamazepine

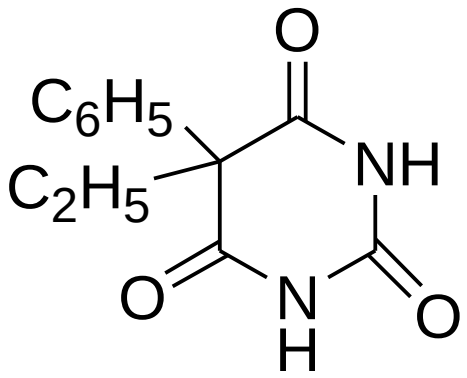


Oxcarbazepine

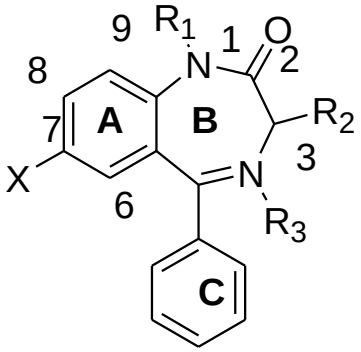
c) Miscellaneous



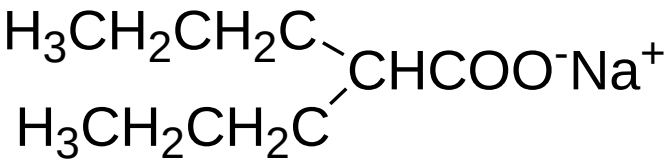
Primidone(deoxyphenobarbital,Prodrug)



Phenobarbital



Benzodiazepines



Sodium Valproic acid

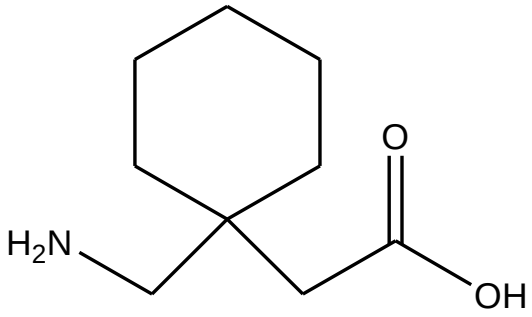
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Newer antiepileptic agents

Gabapentin

-The name & structure again suggest GABA mediated pharmacology

-But surprisingly the mechanism of action remain unknown



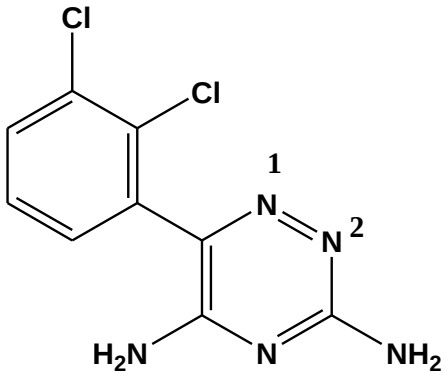
Gabapentin

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Newer antiepileptic agents

Lamotrigine

- Lamotrigine is 1,2,4- triazine compound with antiepileptic profile in experimental models of epilepsy similar to phenytoin & carbamazepine.
- It acts by inhibiting release of excitatory neurotransmitters, especially glutamate

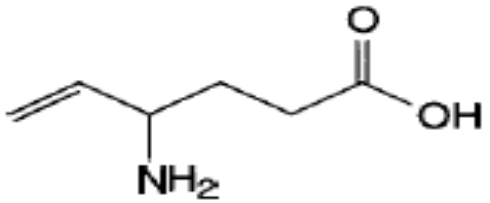


Medchem,Alex T.

Newer antiepileptic agents

Vigabatrin

- Vigabatrin is chemically 4 amino 5 hexanoic acid
- As the name suggest this drug is structural analog of GABA it act as an inhibitor of the enzyme GABA transaminase that degrade GABA thereby enhancing the activity of GABA
- Because of irreversible inhibition the effect last longer than suggested by the half life



3. Tranquilizers

- Give strong sedation without producing sleep and produce a state of indifference and disinterest.
- Effective in reducing excitation, agitation/disturbance, aggressiveness and impulsiveness (major tranquilizers)
- They may be classified into
  - **Major tranquilizers** (neuroleptics)**:** used mainly for treatment of psychosis (schizophrenia and mania)
  - **Minor tranquilizers** (anxiolytics)**:** used to reduce pathological anxiety, agitation and tension

Tranquilizers...

- Dopamine hypothesis of Schizophrenia (and mania) suggests that schizophrenia results from increased dopaminergic neurotransmission
- Hence the strategy is to block these dopaminergic receptors (specifically D_2 and D_3 receptors

Tranquilizers...

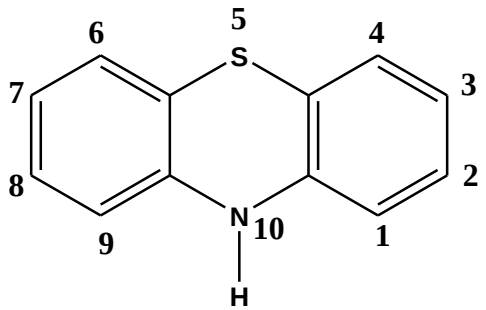
- **Schizophrenia** is a long-term mental disorder of a type involving a breakdown in the relation between thought, emotion, and behaviour, leading to faulty perception, inappropriate actions and feelings, and withdrawal from reality into fantasy and delusion
- **Mania** is a mental illness marked by periods of excitement, delusions, and over activity.

3.1 Major neuroleptics/Antipsychotics

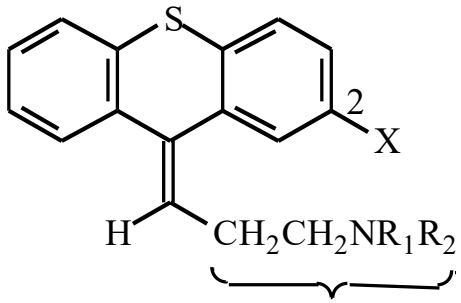
- Are used in the treatment of schizophrenia/mania
  - A disorder involving disruption and disintegration of behavior
- Symptoms
  - Alteration of behavior thought and perception
  - Thought disorder
  - Decreased &/or inappropriate emotional response
  - Auditory hallucination
  - Delusion of being persecuted
  - Withdrawal from social contact

Classification of antipsychotic drugs

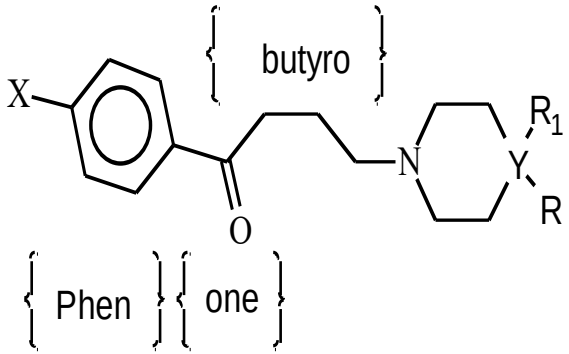
- **Typical Antipsychotic Drugs:**
  - Phenothiazines
  - Thioxanthenes
  - Butyrophenones



Phenothiazine

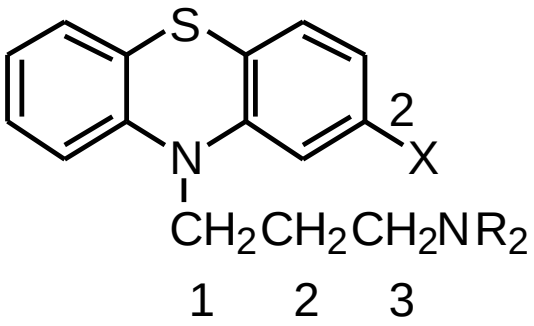


"R"



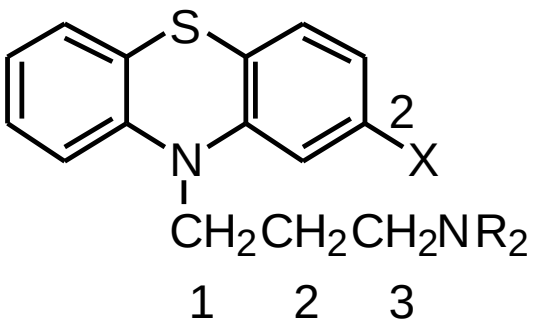
- Most neuroleptics block central D2 receptors in the mesolimbic and mesocortical pathways.
- Newer drugs block both serotonergic and dopaminergic receptors.

Phenothiazine SAR



1.Propyl Side Chain

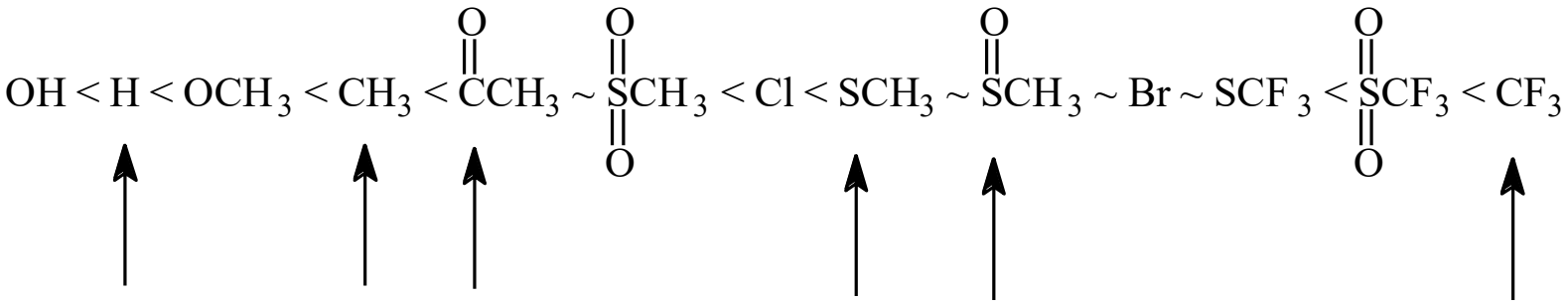
- Propyl is best, butyl is nearly inactive, ethyl has low activity.
- Compounds with ethyl chains often have antihistaminic activity.
- Any substituent at the first position of the side chain decreases activity.
- Substitution of a methyl at position 2 of the chain is OK, phenyl is OK.
- Large aliphatic substituents are not tolerated.
- A larger range of substitutions are tolerated at position 3.
- The nitrogen is often included as part of a ring.



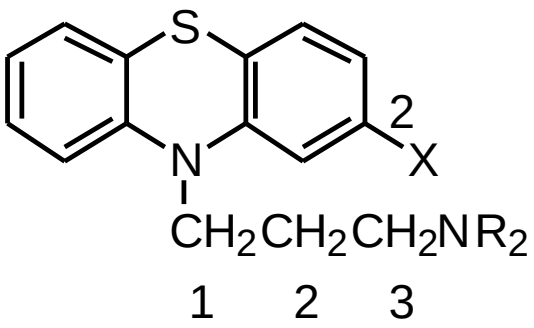
2. Modification of X and the Tricyclic Nucleus

- Highest activity is associated with an electron withdrawing, lipophilic substituent (halogen) at position 2.
- A trend in activity changes can be seen with alterations in the identity of X; increasing lipophilicity and e- withdrawing increases activity.
- Disubstitution of the ring decreases activity, ring cleavage is inactivating.
- Replacing S with C, O, Se, etc. decreases activity.
- Replacing the nitrogen eliminates activity (except in special cases.

Identity of Phenothiazine X and Effects on Potency



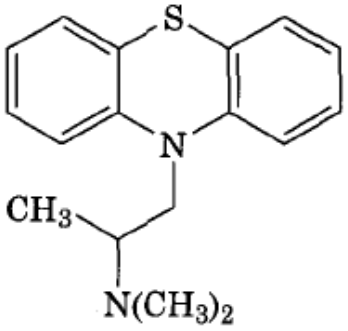
- The order of potency for a “few” X substitutents are shown.
- The most important are indicated by arrows, and Cl of course



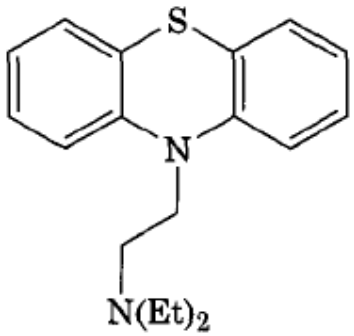
3. Modification of Side Chain Amino Group

- The highest activity is for 3° amines (pKa's of 8-10).
- Methyl R groups on nitrogen have greater activity than larger aliphatic groups.
- The receptor is long and narrow as shown by the amino substitution and by the tolerance to phenyl substitution at C2.
- The amino group can be part of a cyclic structure.
- The cyclic amines include pyrrolidine, piperidine, and piperazine. The piperazine substituent, in particular, generally increases potency.

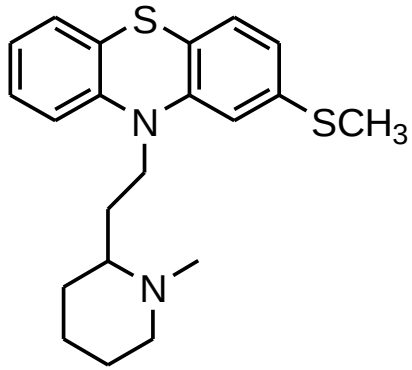
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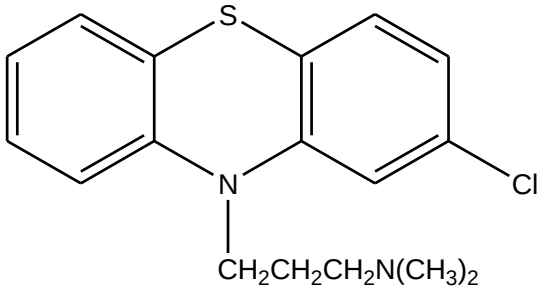
promethazine



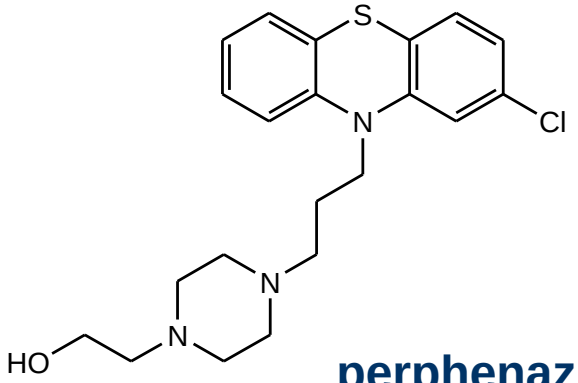
diethazine



thioridazine



chlorpromazine (CPZ)

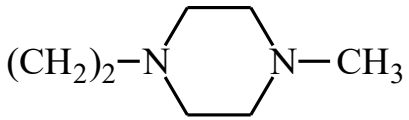


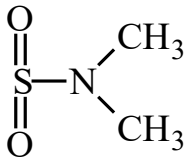
perphenazine

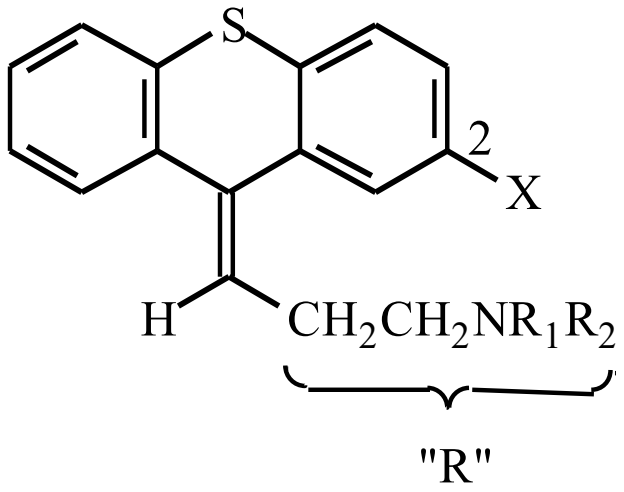
Thioxanthene Antipsychotics

- No ring nitrogen
- Cis double bond compounds are much more active than trans. Double bond reduction reduces activity.
- Other SAR is the same as for the phenothiazines
- Chlorprothixene (Taractan) - 1x chlorpromazine

  R = $(CH_2)_2N(CH_3)_2$ X = Cl
- Thiothixene (Navane) - 10x chlorpromazine

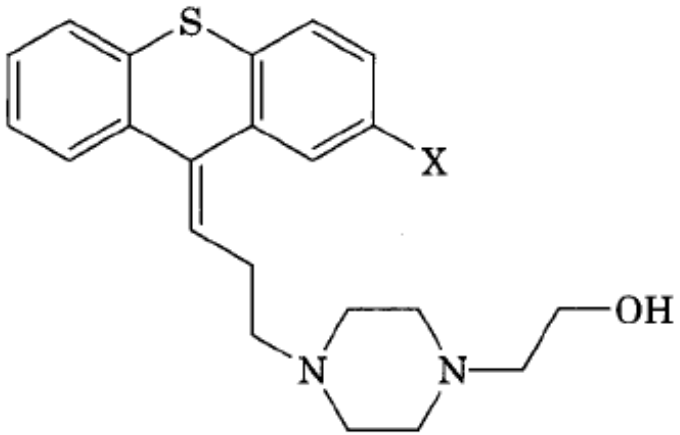
R = 

X = 



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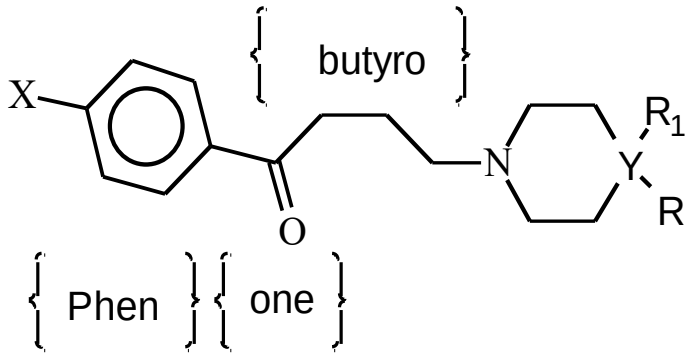
- Examples of thioxanthene derivatives that are potent



X= Cl, clopenthixol
X= CF_3, flupenthixol

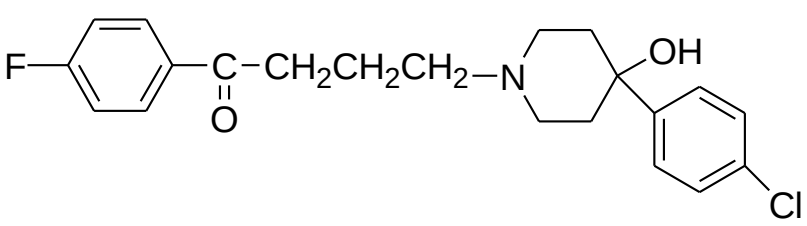
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Butyrophenones

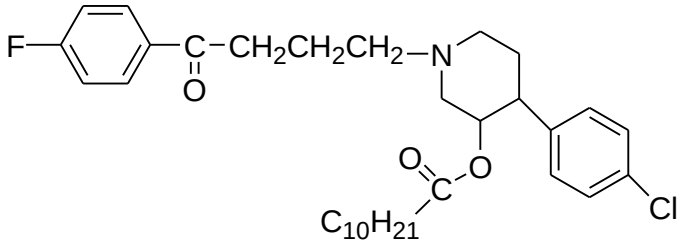


- For highest potency, X is always F. $-OCH_3$ has the lowest potency.
- This tells you that an electron withdrawing group, not electron donating is optimal.
- Shortening, lengthening, or branching of the propyl chain decreases potency
- Replacing the keto oxygen with S, carbon, OH decreases potency
- A 3° N is necessary, analogous to phenothiazines
- Y is usually C with two R groups, but may be N. For example piperazine instead of piperdine.

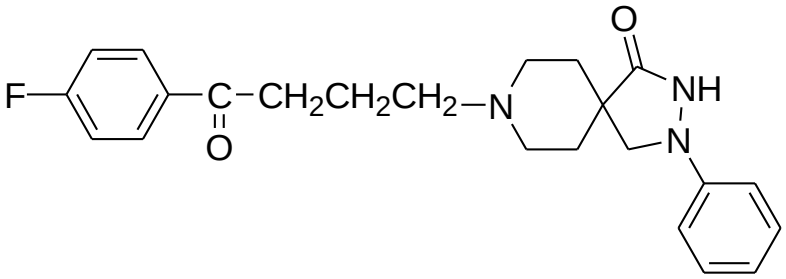
Currently available butyrophenones



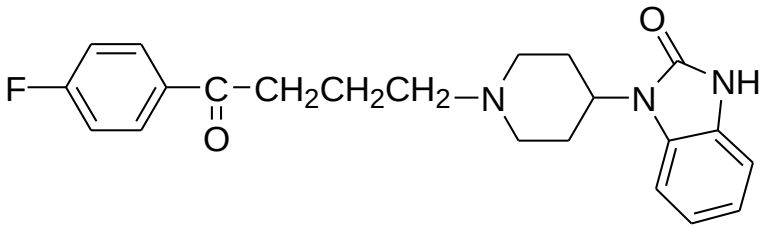
Haloperidol



Haloperidol decanoate



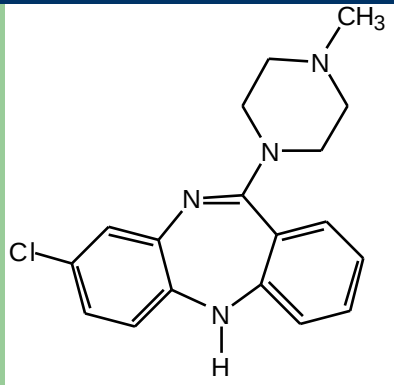
Spirperone



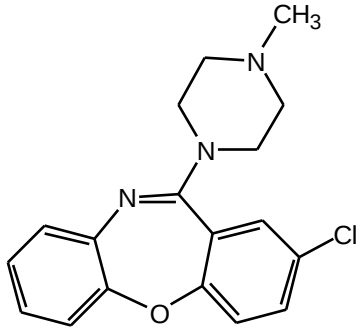
Droperidol

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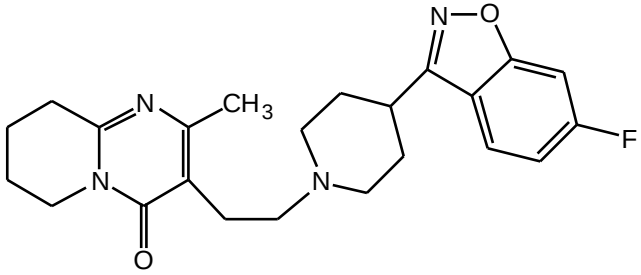
The Atypical Antipsychotics



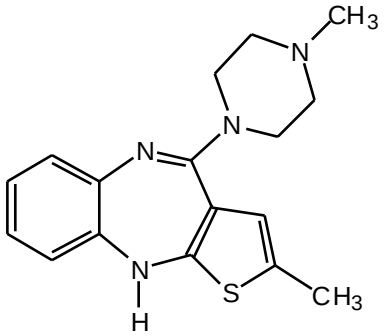
Clozapine



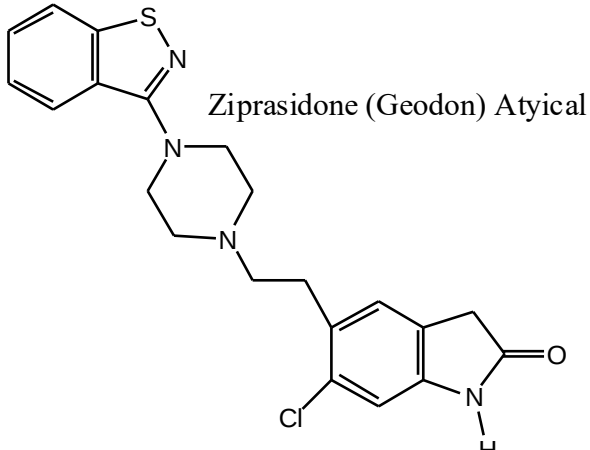
Loxapine



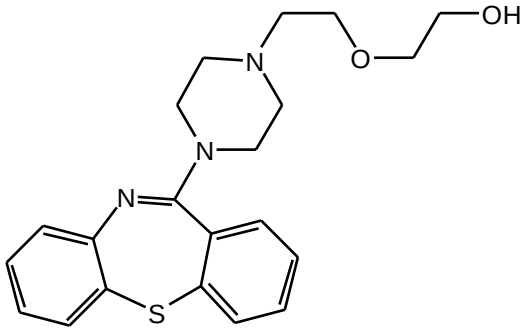
Risperidone (Risperdal) Atypical



Olanzapine (Zyprexa) Atypical



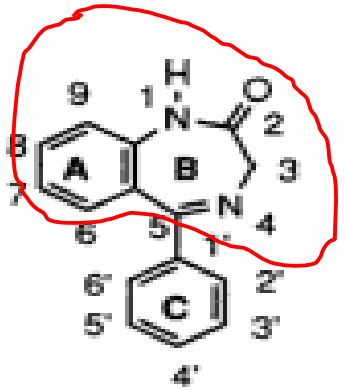
Ziprasidone (Geodon) Atyical



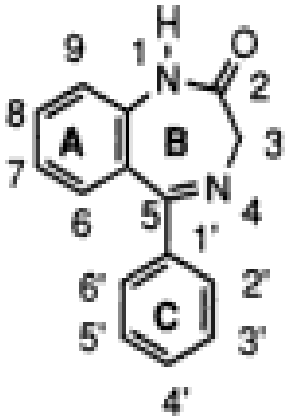
Quetiapine (Seroquel) Atypical

3.2 Minor tranquilizers (anxiolytics) Antianxiety drugs

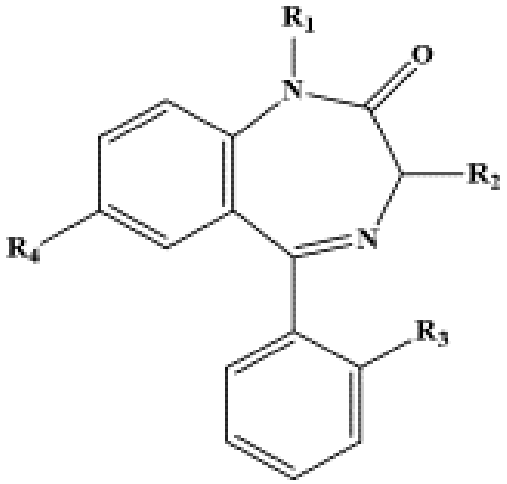
- Anxiety: is sense of apprehensive expectation
  - Part of our daily life
  - Excessive consider as pathological
- The anxious state is a sense of unpleasant anticipation and excessive mental tension.
- Type
  - GAD (chronic and high level of worry)
  - PTSD (Post-Traumatic Stress Disorder)
  - Phobias
  - OCD (Obsessive Compulsive Disorder)



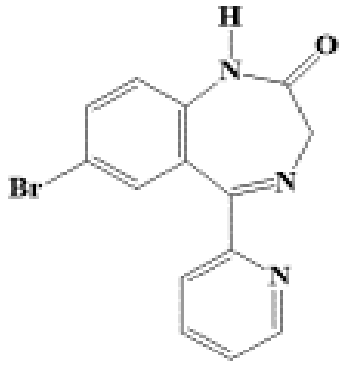
- anxiolytics are drugs used to control moderate or severe anxiety & tension
  - Is situations producing usual
    - Stress
    - Anxiety disorders
- The most used antianxiety drugs are benzodiazepines
  - The name refers to the portion of the structure of benzene fused to the seven member diazepine ring.



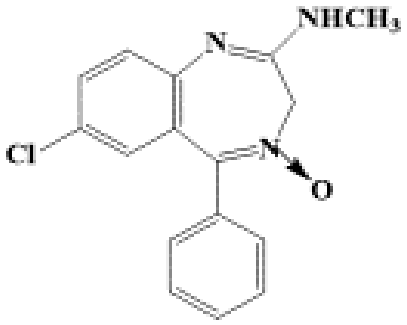
- Benzodiazepines display
  - Sedative hypnotic activity
    - As effective as barbiturates
      - Much safer in terms of their
        - Therapeutic index
        - Addiction
        - Less likely to cause dependence
        - Effect on REM (Rapid Eye Movement) sleep
      - Less drug interaction
        - Thus replacing barbiturates
  - Anti anxiety
  - Anticonvulsant
  - Muscle relaxant effects



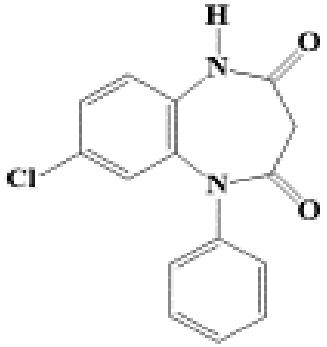
Name of BDZ	R_1	R_2	R_3	R_4
Clonazepam	H	H	Cl	NO_2
Desmethyldiazepam	H	H	H	Cl
Diazepam	CH_3	H	H	Cl
Flunitrazepam	CH_3	H	F	NO_2
Flurazepam	$CH_2CH_2N(C_2H_5)_2$	H	F	Cl
Lorazepam	H	OH	Cl	Cl
Nimetazepam	CH_3	H	H	NO_2
Nitrazepam	H	H	H	NO_2
Norfludiazepam	H	H	F	Cl
Oxazepam	H	OH	H	Cl
Temazepam	CH_3	OH	H	Cl
Clorazepate	H	COOK	H	Cl
Halazepam	CH_2CF_3	H	H	Cl



Bromazepam

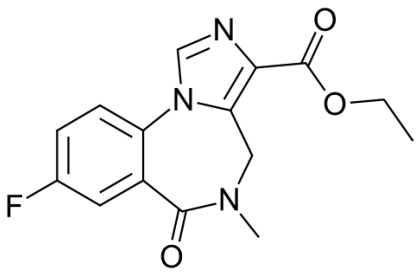


Chlordiazepoxide



Clobazam

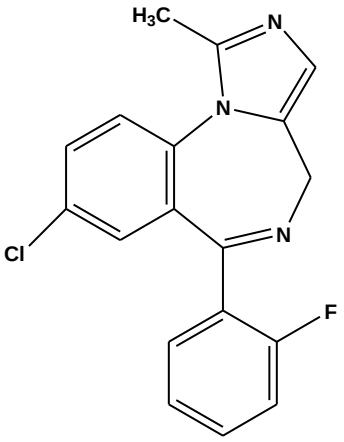
- Compounds with a fused triazole ring (triazolam & alprazolam) & fused imidazole ring (midazolam) were synthesized
- Those compounds are metabolized mainly by hydroxylation of the methyl substituent on the triazolo or imdazolo ring
  - The resulting active hydroxyl compound but undergo quicker conjugation reaction
- Interestingly the presence of an electron attracting at position 7 not required for activity



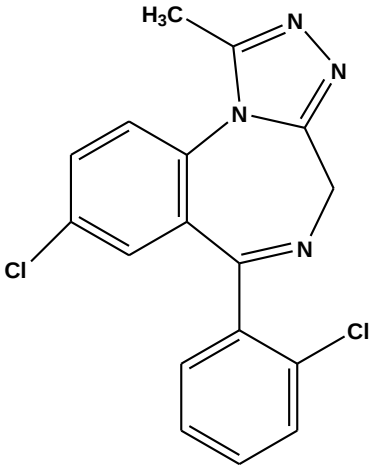
$GABA_A$ receptor antagonist
BZD overdoses

Flumazenil (Anexate)

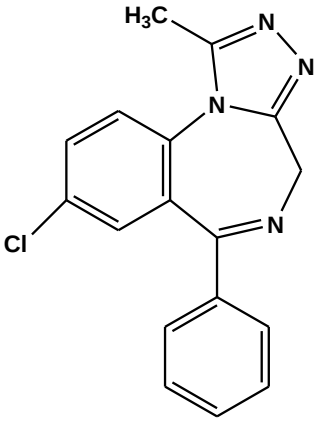
- The triazo moiety increases the drug's binding affinity and stability.
  -As a result, the potency is greatly increased.
- Short half-lives; oxidation of methyl to alcohol and glucuorination
- These compounds have intermediate to short duration of action.
- In these drugs the nitrogen atoms typically are not sufficiently basic to be protonated (ionized) at physiologic pH, but they are sufficiently basic to yield more stable and water soluble salts when treated with strong acids



Midazolam



Triazolam



Alprazolam

- Benzodiazepines have replaced the barbiturates because they have
  - a lower abuse potential,
  - relatively lower adverse reactions (chiefly, death is a relatively common result in barbiturate overdoses) and
  - Lower drug interactions.
  - less tendency to tolerance and dependency

4. Central skeletal muscle relaxants

- Skeletal muscle relaxants are used to decrease muscle spasm or spasticity that occurs in certain neurologic and musculoskeletal disorders.
- Muscle spasm or cramp is a sudden involuntary painful muscle contraction that occurs with trauma or an irritant.

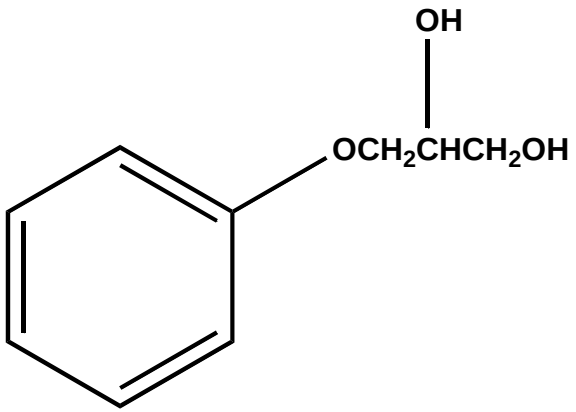
  -Spasms may involve alternating contraction and relaxation (clonic) or sustained contraction (tonic).
- Spasticity involves increased muscle tone or contraction and stiff , awkward movements

  -It occurs with neurologic disorders such as spinal cord injury and multiple sclerosis

- The muscle relaxants may be divided into only two groups
  - Centrally acting muscle relaxants,
    - ✓ Which appears to act on the central nervous system
  - Peripherally acting muscle relaxants.
    - ✓ at the level of the nerve muscle connection

- Spasmolytic drugs are diverse in their chemical structures and their site and mechanism of action
- The first drug recognized to exhibit spasmolytic activity was antodyne or 3- phenoxy- 1,2- propanediol

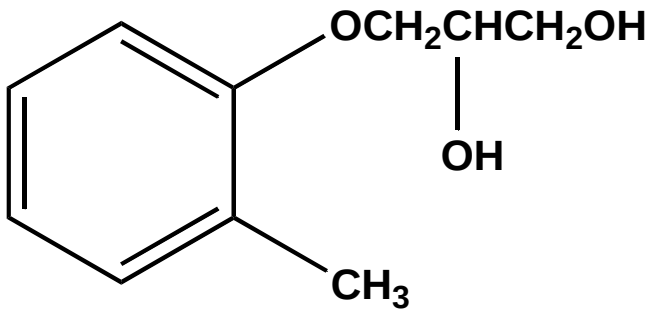
-The duration of its skeletal muscle relaxant effect however was to short lived to be clinically useful.



- SAR studies of a series of simple glyceryl ethers related to antodyne, lead to development & introduction of mephenesin

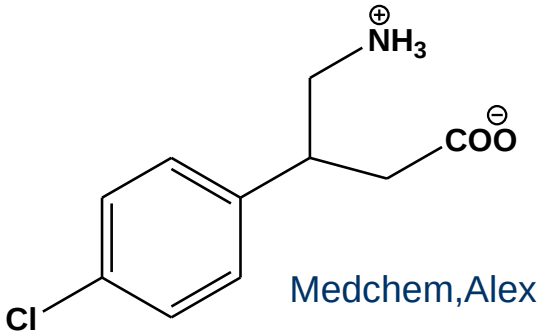
  -Is a prototype of intraneuronal blocking type of muscular relaxant

  -Because of its safety and selective action on the spinal cord



Baclofen

- In attempt to elevate GABA activity in the brain more directly, the GABA analog baclofen was developed
  - -In contrast to GABA baclofen is more lipophilic
  - -Complete absorption following oral administration
  - -Penetrate BBB
- Exert its action by stimulating $GABA_B$ receptor subtypes
  - -Resulting in hyper polarization



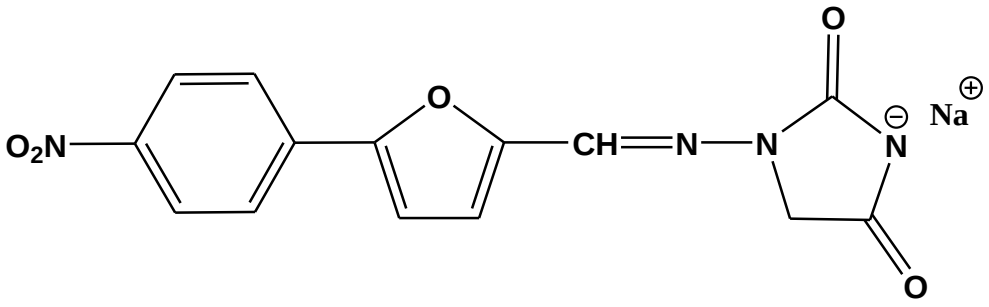
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Dantrolene

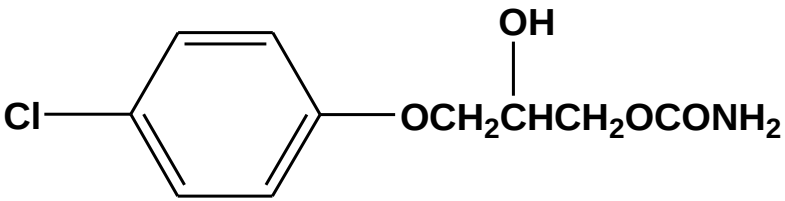
- Bind to a calcium channel protein (called the ranodine R) on sarcoplasm reticulum to close channel & inhibit release of calcium
- Is a hydantoin derivative (are weak bases, *PK*a = 7.5)

  -Cross BBB

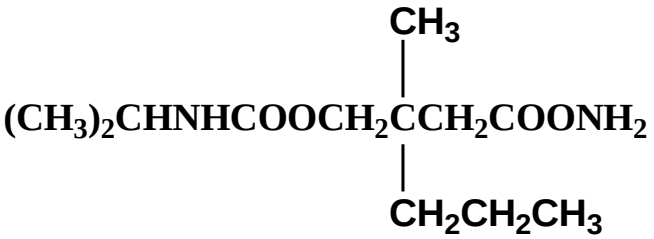
  -Associated CNS side effects



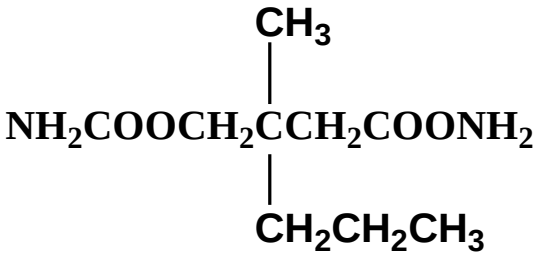
Medchem,Alex T.



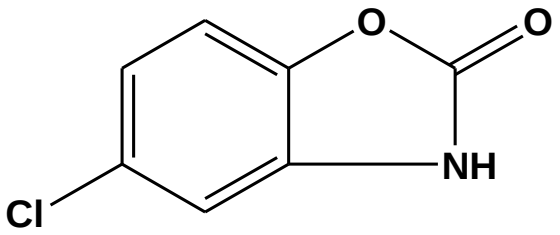
Chlorphenesin



Carisoprodol



Meprobamate



Chlorzoxazone

B. CNS Stimulant drugs

1. Analeptics

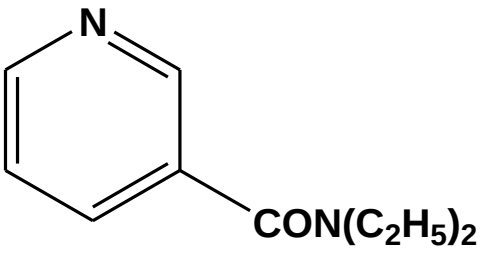
2. Antidepressants

3. CNS adrenergics

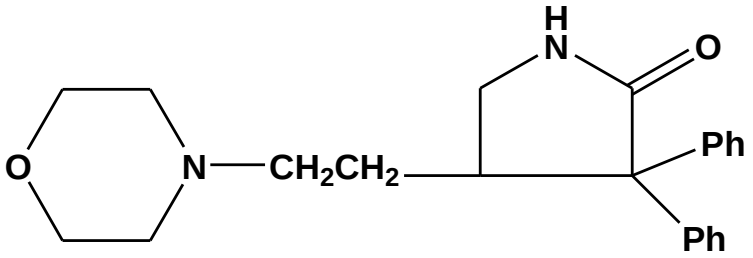
4. Xanthine derivatives

1. Analeptics

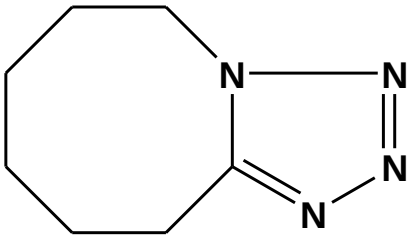
- Analeptics counteract narcosis, with a specific stimulant action on the central nervous system.
- These are primarily employed to combat the drug-induced respiratory depression.
- An excessive dose of analeptics may result a wide-spread stimulation of the brain that may ultimately cause convulsions.
- Agent that cause nonselective CNS stimulation and causes muscle contraction
- Particularly a term used to describe compounds that cause contraction and rigidity of the muscles of respiration



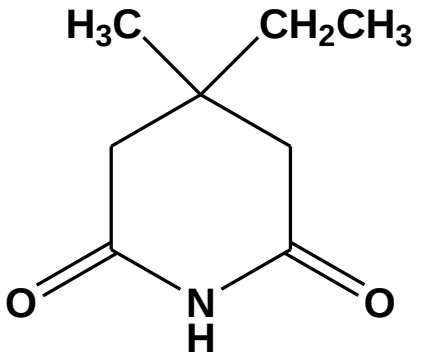
Nikethamide



Doxapram



Pentetrazol



Bemegride Medchem,Alex T.

2. Antidepressant

- Depression is due to low level of monoamines (NE, serotonin & dopamine) in the synapse
- It is characterized by
  - Anhedonia (lack of pleasure)
  - Sadness
  - Helplessness
  - Inferiority
  - hopelessness,



- Worthlessness
- Crying
- Suicidal tendencies
- fatigue or decreased energy
- restlessness, irritability
- sleep disturbances



- Monoamine reuptake inhibitors and monoamine oxidase inhibitors:
  - Act as antidepressant

    History of antidepressants
- Most antidepressant drugs were discovered by serendipity
- Imipramine, the first TCA, develop during the search of drugs for schizophrenia

Theory of depression

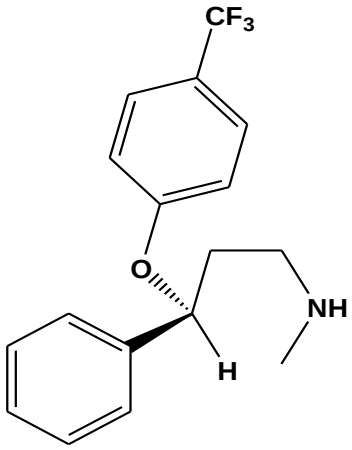
- Monoamine hypothesis suggests that small amount of neurotransmitter causes depression
  - Drugs acts by normalizing this deficit.
- The theory led to the development of the next major class of antidepressants, the selective serotonin reuptake inhibitors (SSRIs).
  - Those drugs selectively inhibit the reuptake of serotonin and there by increase serotonin activity in the brain.

Classification of Antidepressant

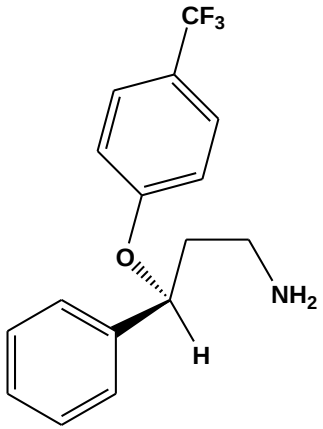
1. Selective serotonin reuptake inhibitors (SSRIs)
2. Selective norepinephrine reuptake inhibitors (SNRIs)
3. Norepinephrine and serotonin reuptake inhibitors (NSRIs)
4. Norepinephrine and dopamine reuptake inhibitors (NDRIs)
5. Monoamine oxidase inhibitors (MAOIs)

Selective serotonin reuptake inhibitors (SSRIs)

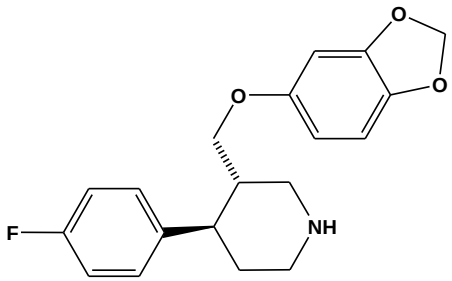
- SSRIs represent a relatively new class of antidepressant dugs.
- 5-HT (serotonin) is a major player in depressive illness, and serotonergic pathways are closely related to mood disorders, especially depression.
- Thus drugs affecting the 5-HT levels in the neural synapse and serotonergic pathways may lead to effective therapy of depression



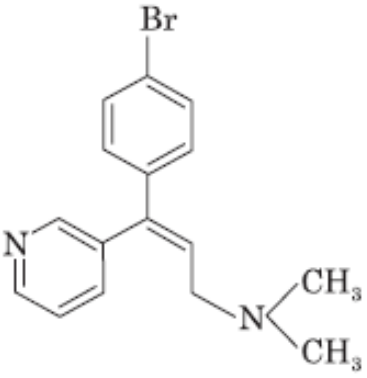
Fluoxetine



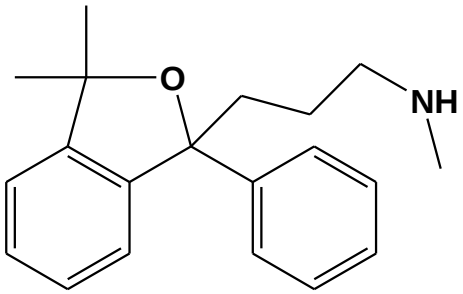
Norfluoxetin



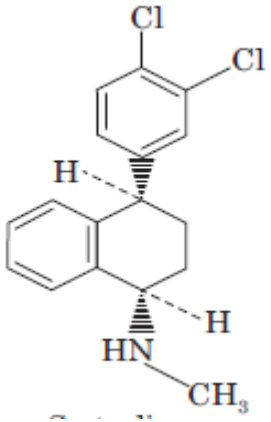
Paroxetine



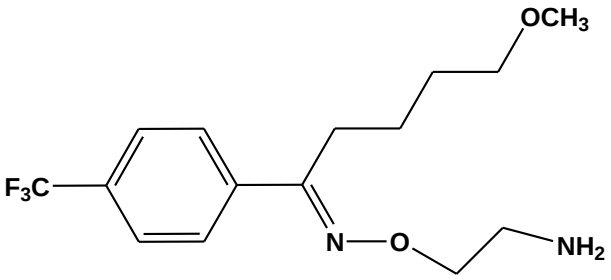
Zimelidine



Talopram



Sertraline



Fluvoxamine

Norepinephrine reuptake inhibitors (SNRIs)/TCA

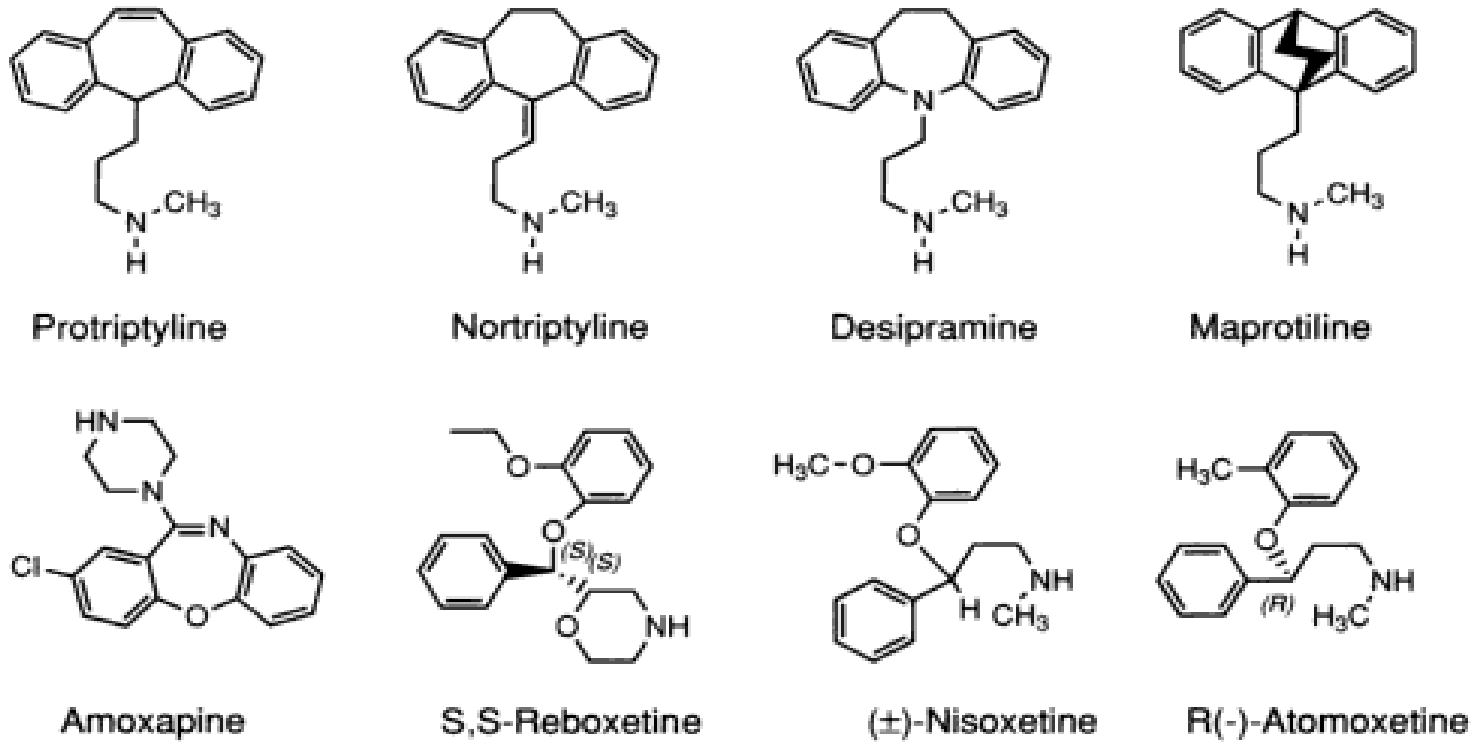
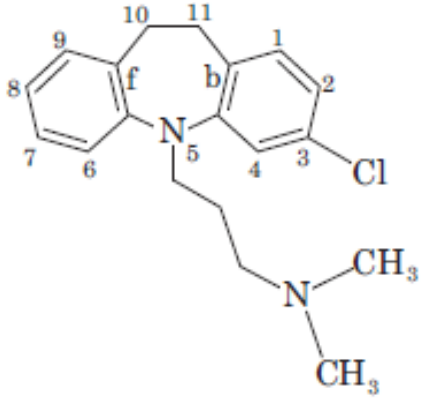


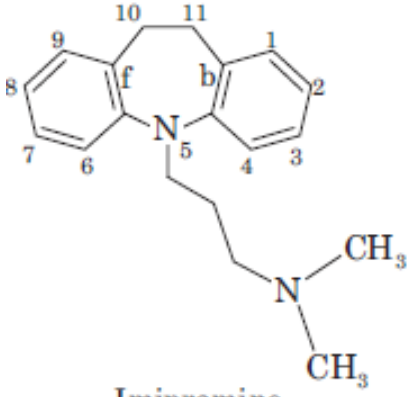
Fig. 21.8. Selective norepinephrine reuptake inhibitors (SNRI).

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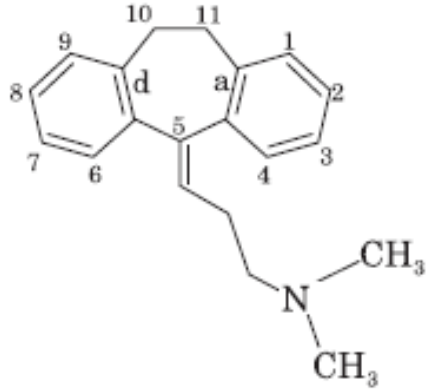
Norepinephrine and serotonin reuptake inhibitors (NSRIs)/TCA



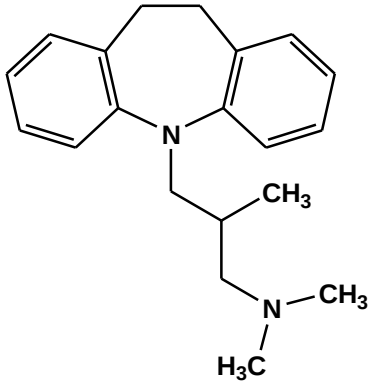
Chlorimipramine



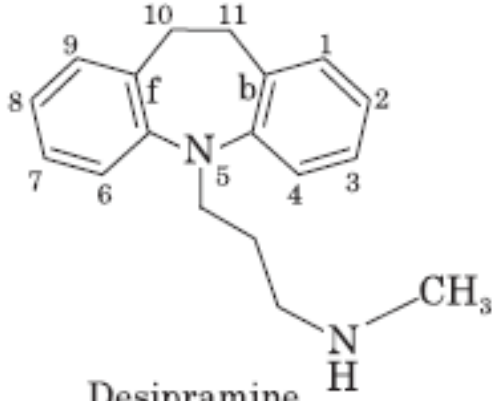
Imipramine



Amitriptyline

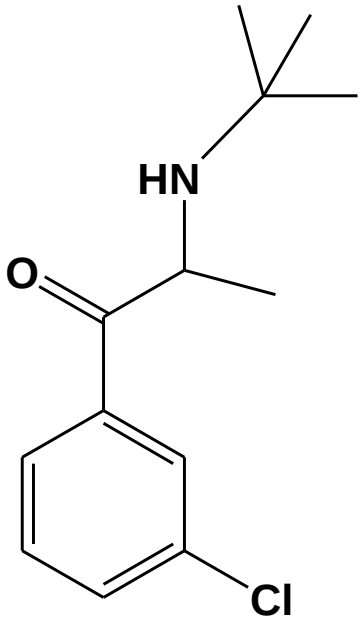


Trimipramine

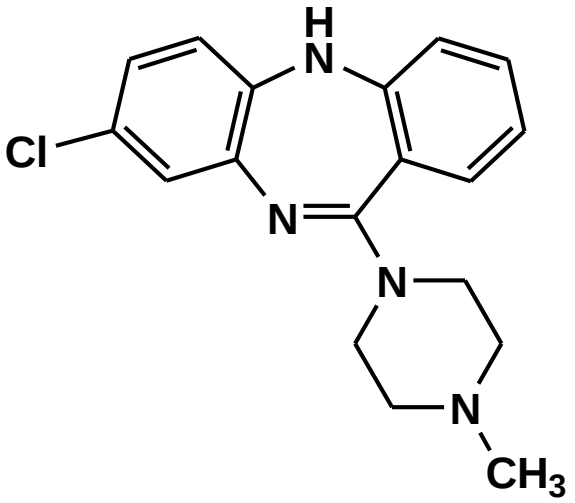


Desipramine

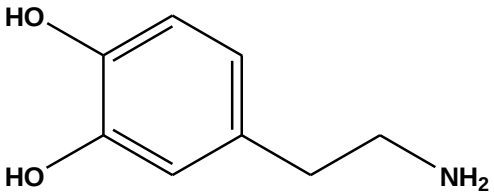
Norepinephrine and dopamine reuptake inhibitors (NDRIs)



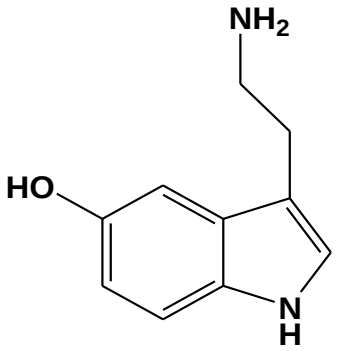
Bupropion



Clozapine

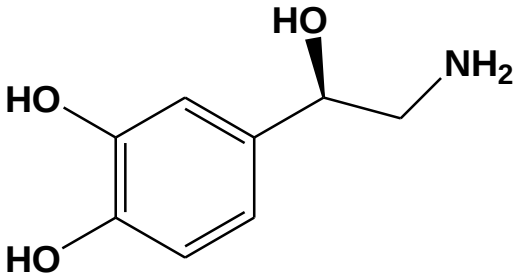


dopamine



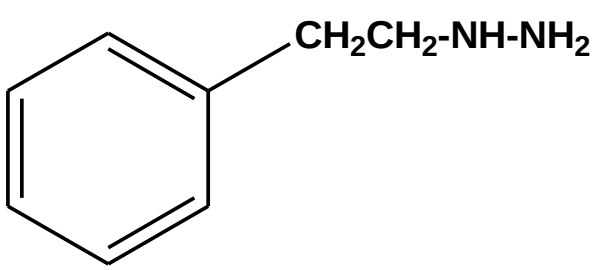
Monoamine oxidase inhibitors (MAOI)

- Monoamine oxidase catalyses oxidative deamination of biogenic amines
  - Norepinephrine
  - Serotonin and Dopamine

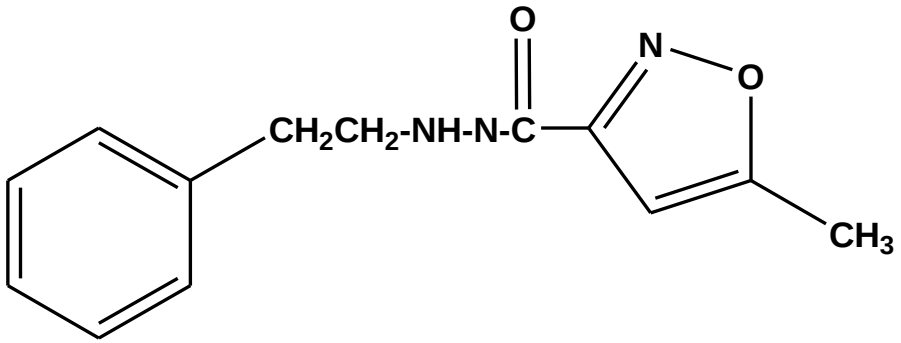


- MAOIs work by blocking the intra- neuronal oxidation & deamination of brain biogenic amines
- Higher concentration of the neurotransmitters stored in the vehicle
  - Available for the release on demand

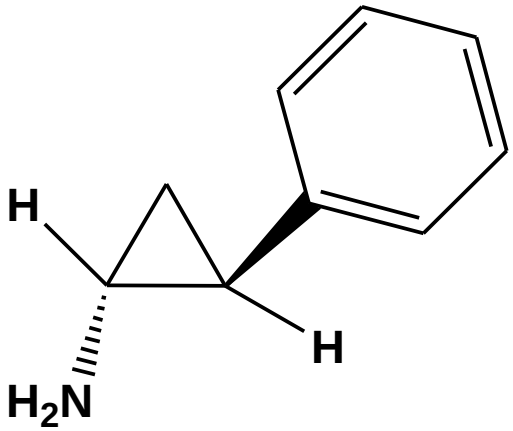
- MAOIs are antidepressants, which are efficacious in the treatment of atypical depression (vegitative symptoms, anxiety, initial insomnia), panic disorder, anxiety and recently discovered borderline personality disorder.
- MAO inhibitors are especially useful in treating people whose depression is combined with other problems such as anxiety, panic attacks, phobias, or the desire to sleep too much.



Phenelzine (Nardil)

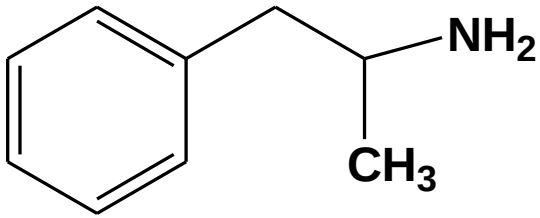


Isocarboxazide (Marplan)

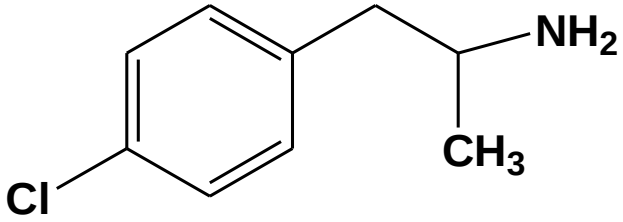


Tranylcypromine

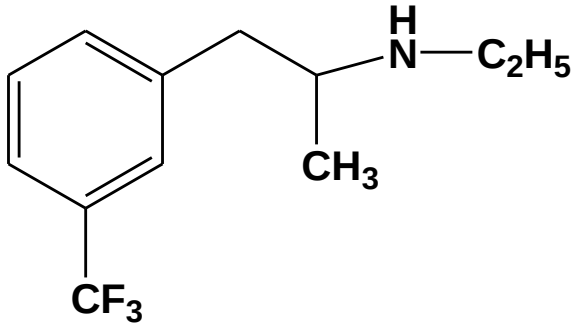
3. CNS Adrenergics



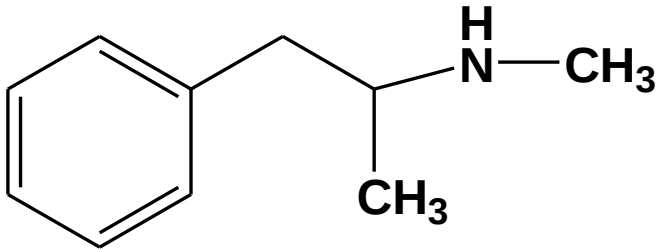
Amphetamine



Chloroamphetamine

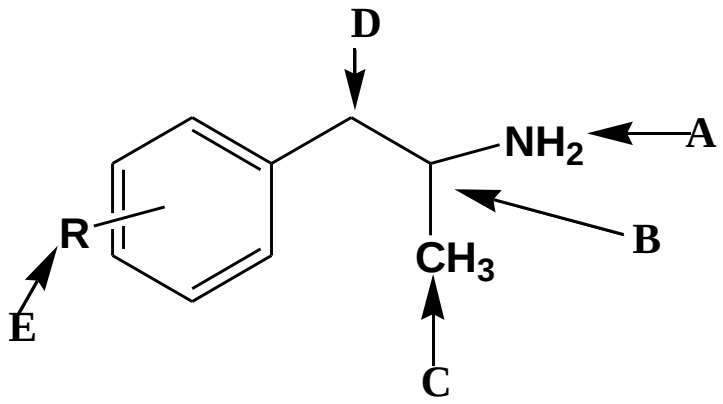


Fenfluramine



Methamphetamine

Amphetamine SAR

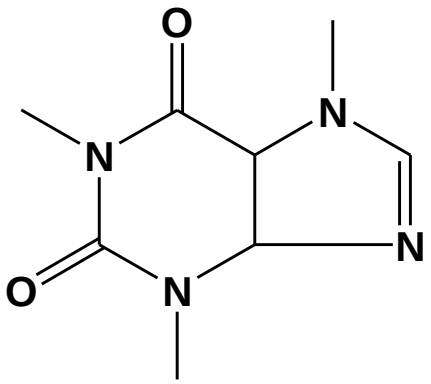


Position	Amphetamine-like action
A: Terminal amine	N-Methyl > NH_2 > NHR > NR_2
B: Chiral center	S(+) > (±) > R(-)
C: α-Methyl group	Homologation decreases potency Replacement by H decreases potency
D: β-Position	β-OH: reduces potency β =O: retains activity and potency
E: Aromatic substitution	Unsubstituted aromatic ring prefered

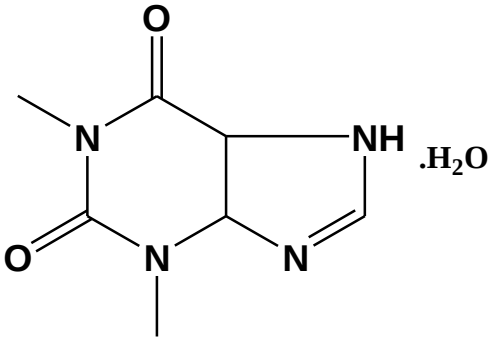
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4. Xanthine derivatives

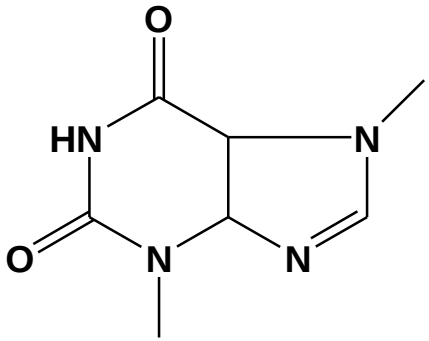
- The xanthine derivatives have a wide spectrum of therapeutic applications ranging from their stimulation of cardiac muscle, enhanced diuresis, stimulation of CNS and finally a soothing relaxation of bronchi and the coronary arteries.
- A few typical examples of the members of this category are : caffeine, theophylline, theobromine, aminophylline, etofylline and proxyphylline.



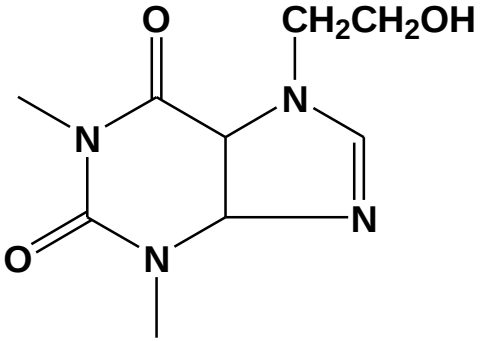
Caffeine



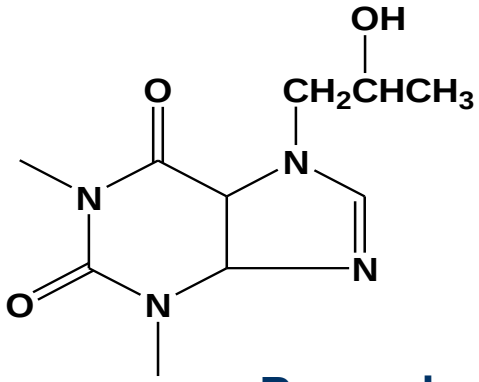
Theophylline



Theobromine



Etofylline



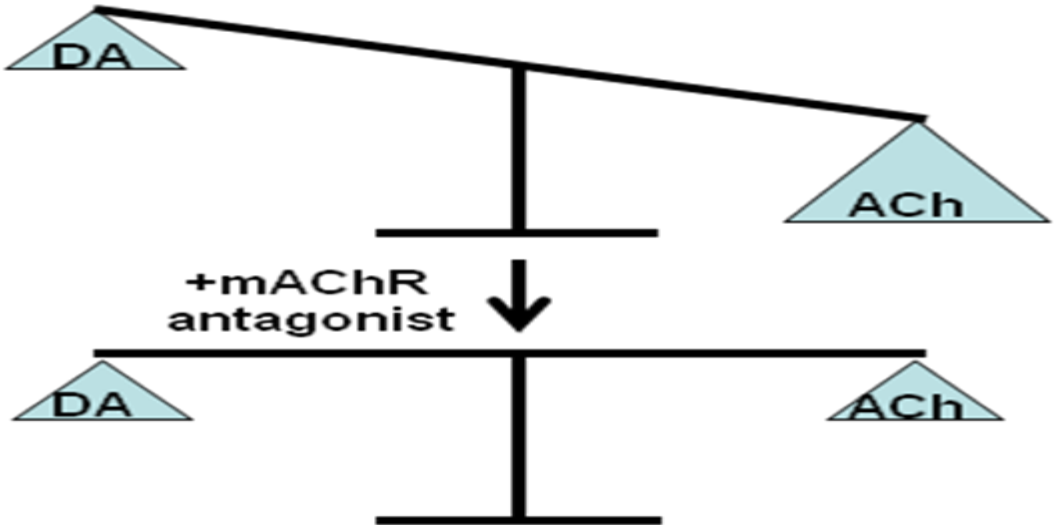
Proxyphylline

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Anti-parkinsonian agents

- Parkinson disease is a slowly progressive degenerative neurologic disease resulting in lack of control over movement, poor balance and coordination
  - ✓ Tremor,
  - ✓ rigidity,
  - ✓ sluggish neuromuscular response and
  - ✓ postural instability
- In Parkinsonism dopaminergic input is deficient and chloinergic output remain unchanged

- Parkinson’s disease, a disabling motor impairment disorder due to the loss of nigrostriatal dopamine
- The loss of dopaminergic inhibition permits Cholinergic hyperactivity in the brain centre



- The antiparkinsonian agents are thus either

  -Dopaminergic

  or

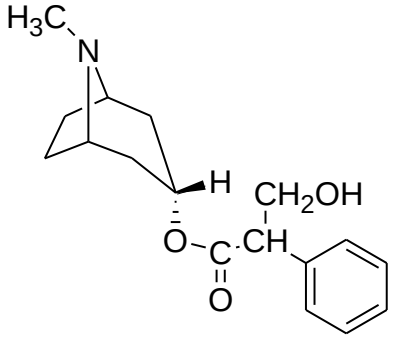
  -Anticholinergic

Anticholinergic antiparkinsonian agents

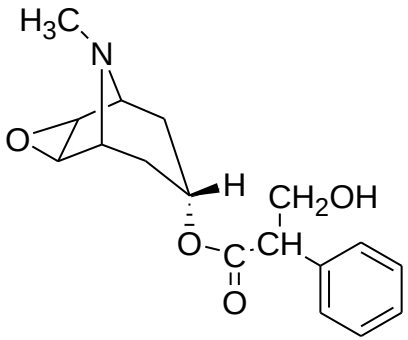
- Atropine & scopolamine were used successfully
  - They block excitatory cholinergic influence in the basal ganglia
- Newer established anticholinergic agents include
  - Benzotropine
  - Trihexyphenidyl
  - Procyclidine
  - Orphenadrine

Anticholinergics….

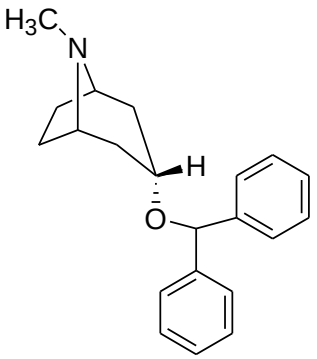
- Anticholinergics used



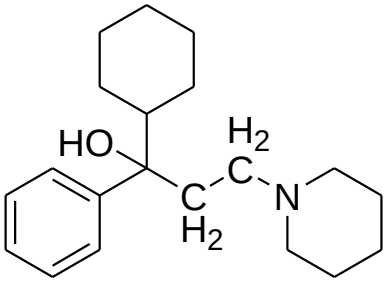
Atropine



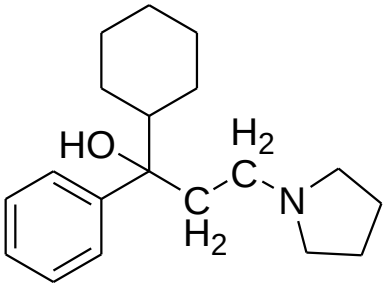
Scopolamine



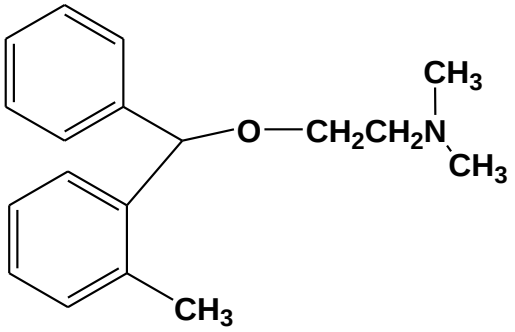
Benzotropine



Trihexyphenidyl



Procyclidin



Orphenadrine

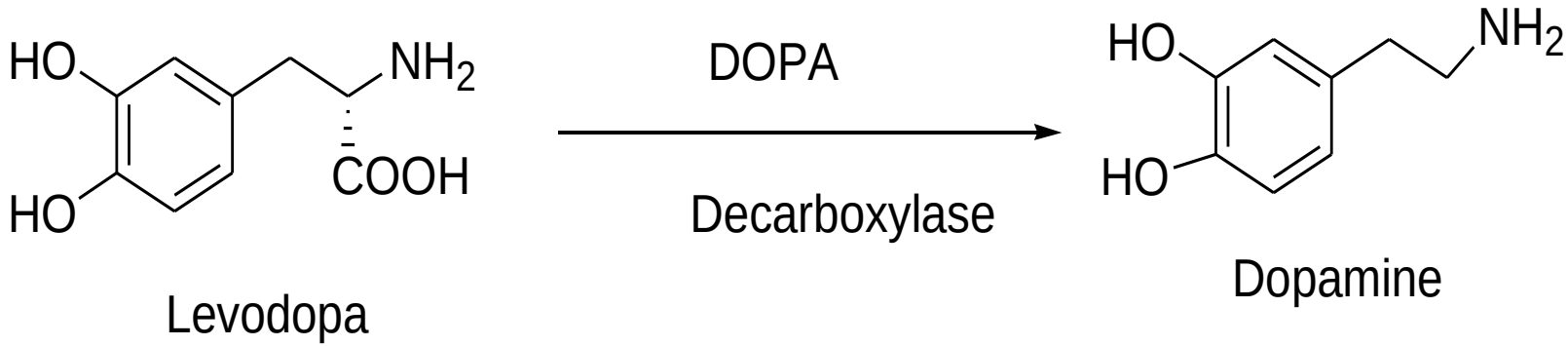
Dopaminergic therapy

- Several approaches are used to eliminate dopamine deficiency in striatum
  - Augmentation of brain synthesis of dopamine
  - Presynaptic dopamine release stimulation
  - Direct stimulation of dopamine receptor
  - Inhibition of dopamine metabolism

Dopaminergic …

1. Augmentation of dopamine synthesis

- **Levodopa therapy**

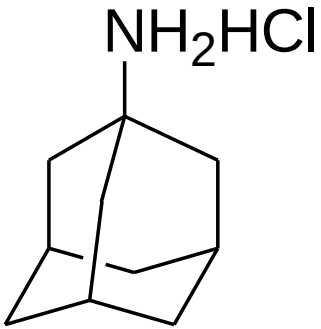


Levodopa

Dopamine

2. Stimulation dopamine release

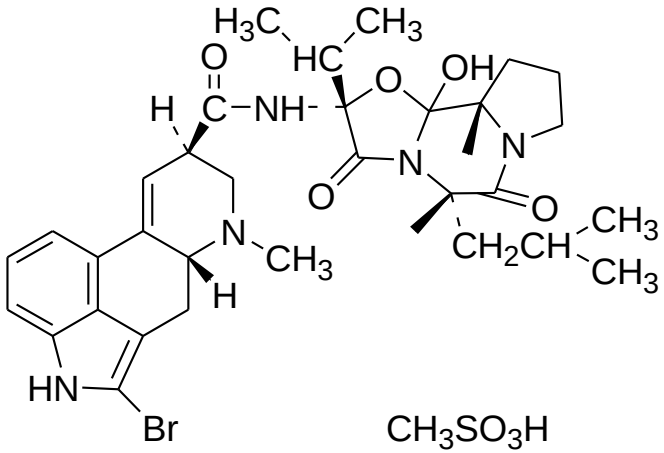
- Agents that release dopamine from neuronal storage can be used
- Amantidine is an example of such drug



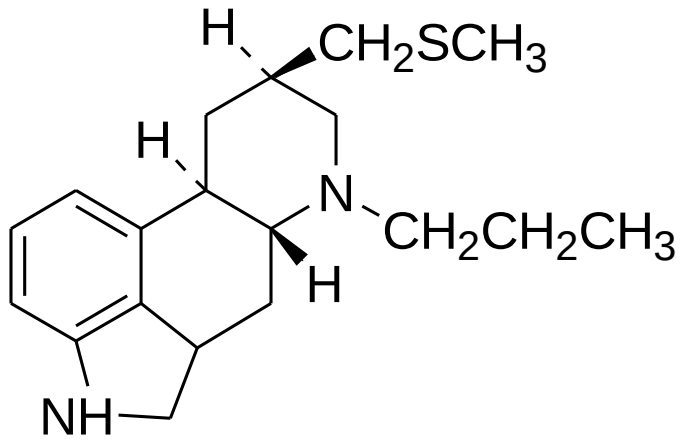
- Amantadine increase dopamine levels at postsynaptic receptor by
  - Decreasing pre-synaplic reuptake
  - Enhancing synthesis & release
- The cage structure makes it lipophilic & passes BBB

3. Direct Dopamine agonist

- Includes the ergot alkaloid derivatives **Bromocriptine** and **Pergolide**
- Dopamine agonists have the advantage in that their effect is independent of striatonigral degeneration

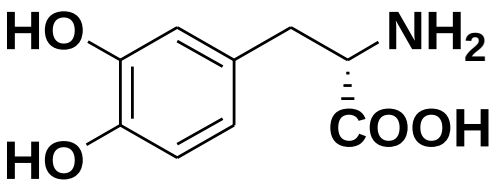


Bromocriptine



Pergolide

4. Inhibition of metabolism



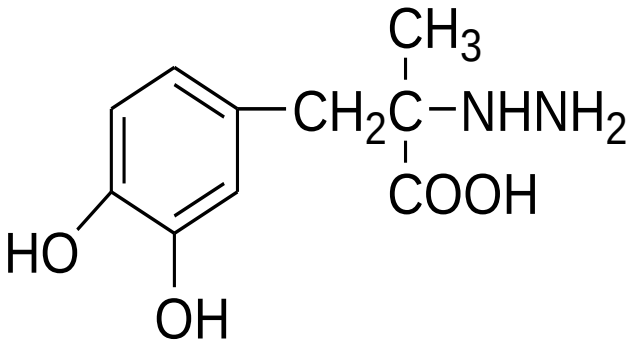
- L-dopa is prodrug for dopamine
- Is given orally
- Only small fraction reaches the brain
- Dose and peripheral side effect decreased
  - By preventing decarboxylation of L- dopa outside CNS

-Is achieved by using DOPA decarboxylase inhibitors that do not cross BBB

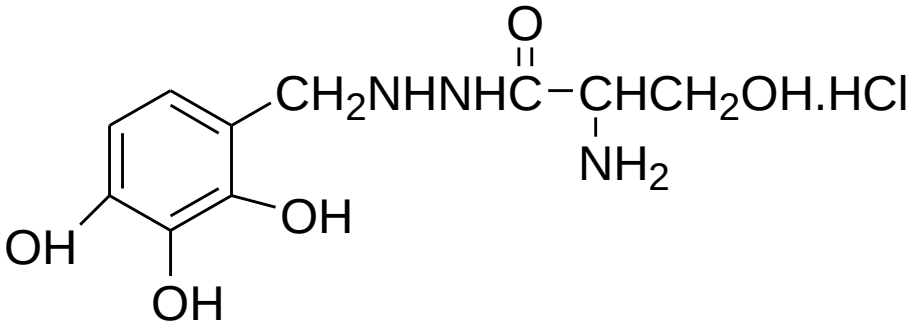
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- Inhibitors such as

  -Carbidopa & benserazide given with L- DOPA



Carbidopa



Benseriazide hydrochloride

- This allow administration of small dose
- Prevent metabolic formation of peripheral dopamine

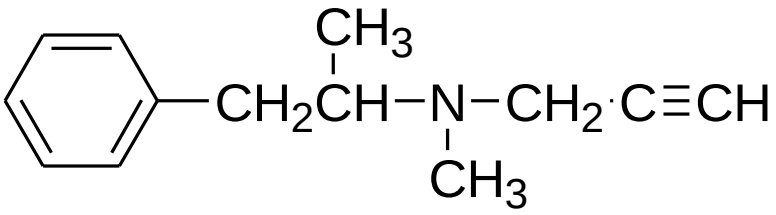
  - less L- DOPA is broken down outside CNS and more dopamine product at the site where it is needed
- The inhibitors are

  -Polar

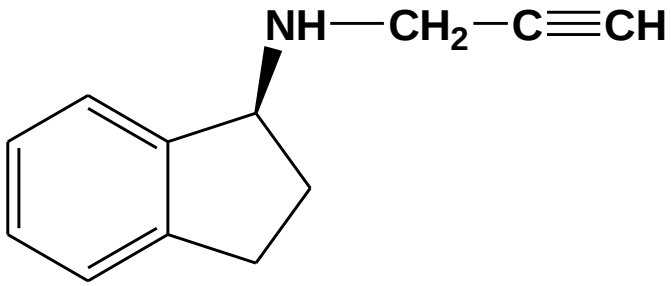
  -Ionic at physiological PH

  - can not cross BBB

- Another means of increasing dopamine level is inhibition of MAO-B
- Selegiline is selective inhibitor of MAO type B and prevents breakdown of dopamine selectively



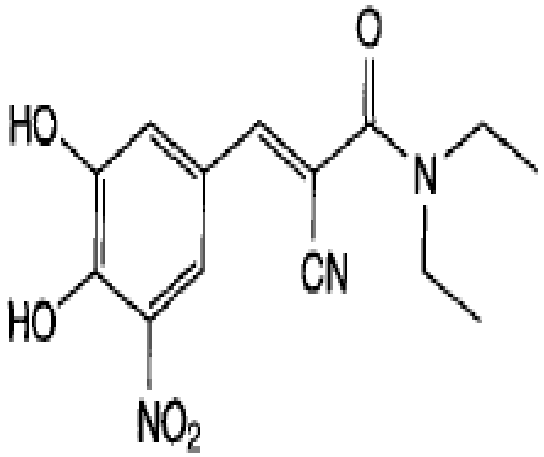
Selegiline



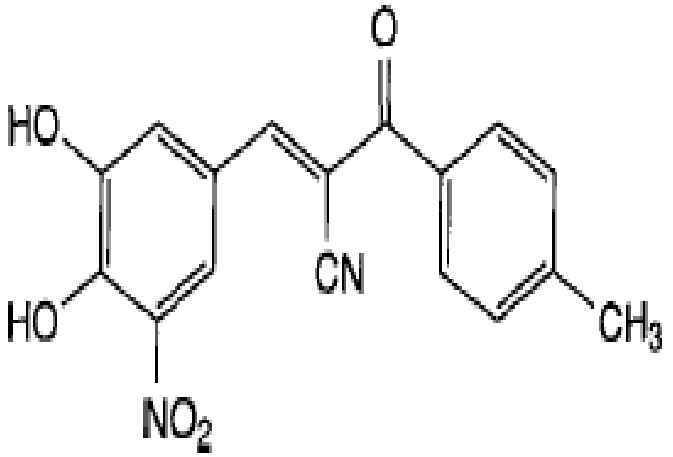
Rasagilne

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- **Catechol-O-Methyltransferase Inhibitors**
  - They potentiate L-dopa and prolong its actions by inhibiting the metabolic inactivation of L-dopa and DA.



Entacapone



Tolcapone

Narcotic Analgesics & Antagonists

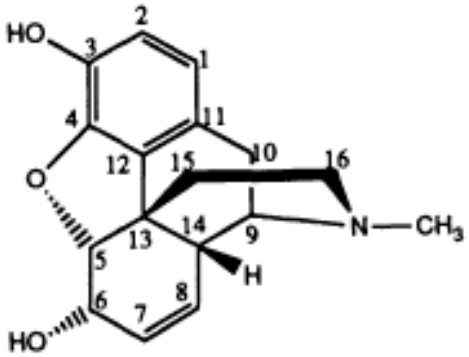
- The term 'opium alkaloids' has been used to cover all narcotic analgesics, whether they are synthetic compounds, partially synthetic, or extracted from plant material.
- extracted from opium—the sticky exudate obtained from the poppy *(Papaver somniferum)*

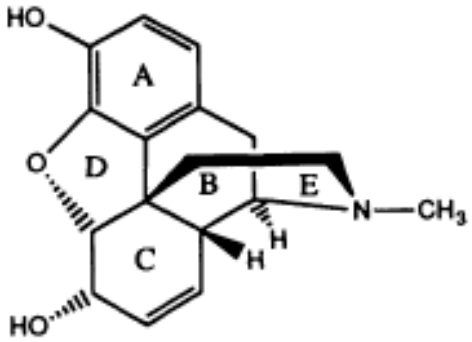


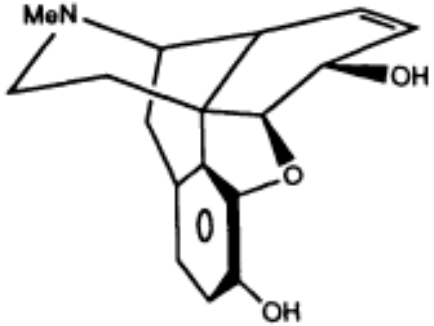
Narcotic is an addictive drug, especially an illegal one, affecting mood or behaviour.

Morphine

Structure and properties







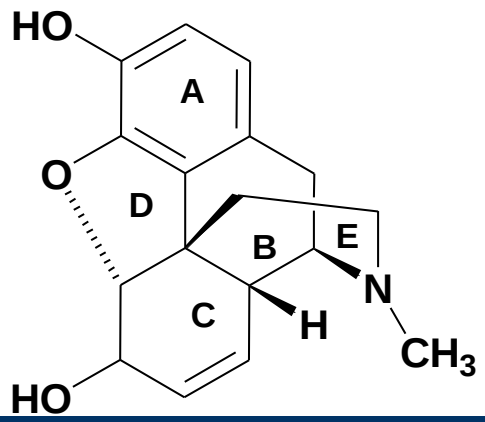
- Morphine is still one of the most effective painkillers available to medicine.
- It is especially good for treating dull, constant pain rather than sharp, periodic pain.

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Morphine…

- Unfortunately, it has a large number of side-effects which include the following:
  - **depression of the respiratory centre**
  - constipation
  - excitation
  - euphoria
  - nausea
  - pupil constriction
  - **tolerance**
  - **dependence**

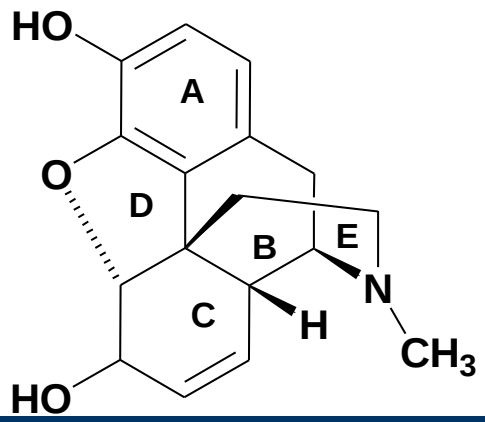
Morphine….



Structure-activity relationships

- The molecule contains five rings labelled A-E and has a pronounced T shape.
- It is basic because of the tertiary amino group, but it also contains a phenolic group, an alcohol group, an aromatic ring, an ether bridge, and a double bond.

SAR…

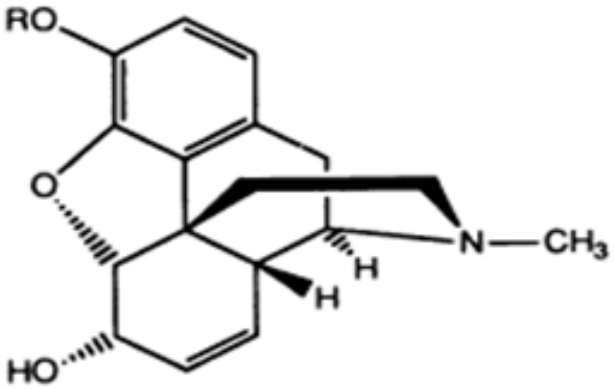


i. Changes which do not affect the basic skeleton of the molecule

The phenolic OH

- By methylating the phenolic OH, the analgesic activity drops drastically and codeine is only 0.15 percent as active as morphine.
- This drop in activity is observed in other analogues containing a masked phenolic group.
- Clearly, a free phenolic group is crucial for analgesic activity.

SAR…



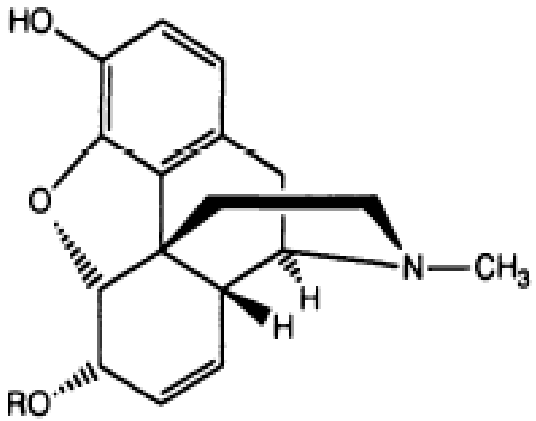
R = Me CODEINE
R = Et ETHYLMORPHINE
} Analgesic Activity ↓

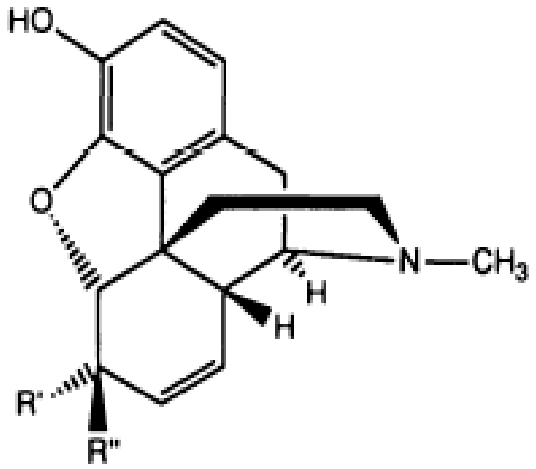
- Codeine a prodrug for morphine, it is metabolized in *vivo* to morphine when it is going to be given orally
- Etherification reduces the analgesic activity but esterification increases activity because of increased lipophilicity and distribution to the CNS

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SAR…

- The 6-alcohol





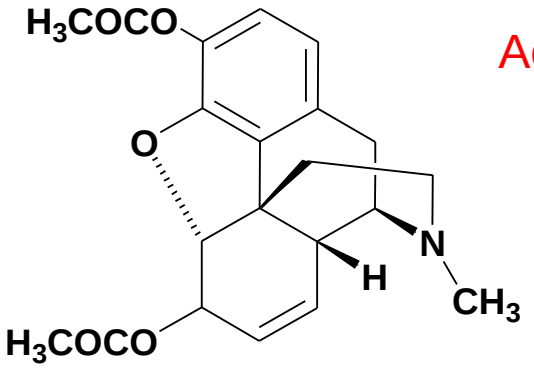
R		Analgesia wrt morphine
Me	Heterocodeine	5x
Et	6-Ethylmorphine	greater
Acetyl	6-Acetylmorphine	4x

R'	R"	Analgesia wrt morphine
H	OH	Increased
H	H	or
Ketone	Ketone	similar

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SAR…

Heroine



Activity:
2x that of morphine b/c polar groups are hidden, making it easy to cross BBB.

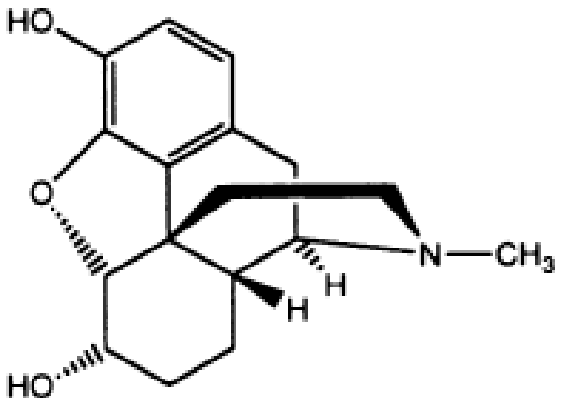
- General observation;
  - masking or the complete loss of the alcohol group does not decrease analgesic activity and, in fact, often has the opposite effect.
  - improvement in activity is due to the pharmacokinetic properties of these drugs rather than their affinity for the analgesic receptor.
  - In other words it increase the amount of the drug that will reach to the CNS
    - Example compare the activities of morphine, heroine and 6 acetyl morphine.

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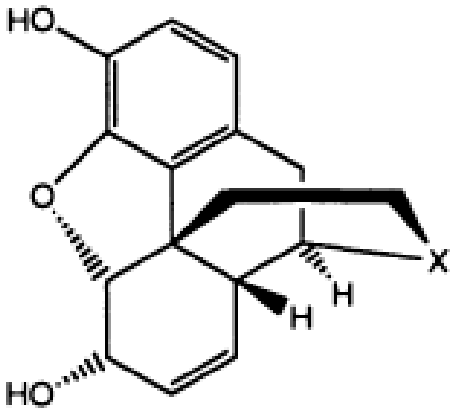
The double bond at 7-8

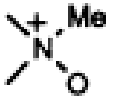
- Several analogues including dihydromorphine (below) have shown that the double bond is not necessary for analgesic activity.



SAR…

The N-methyl group



X		Analgesic Activity wrt morphine
NH	Normorphine	25%
	N-Oxide	0%
	Quaternary salt	0%

- The methyl group is not must but the nitrogen should be there and it must be ionized.

SAR…

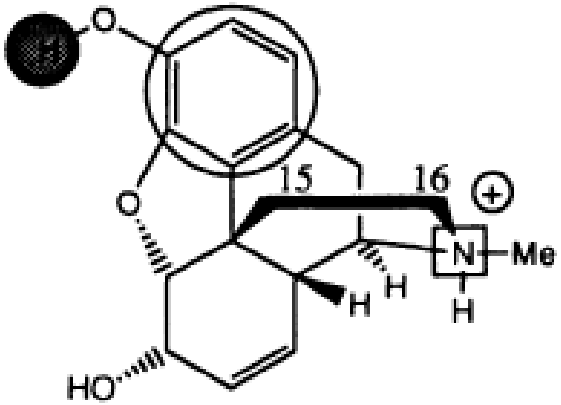
The aromatic ring

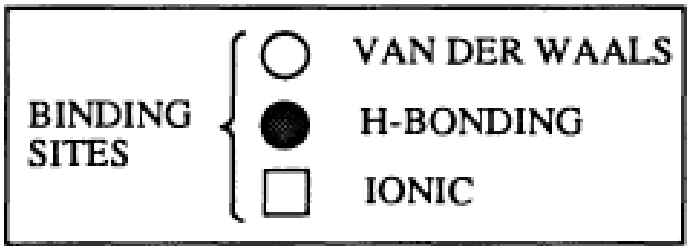
- The aromatic ring is essential. Compounds lacking it show no analgesic activity.

The ether bridge

- The ether bridge is not required for analgesic activity.

➢ To sum up, the important functional groups for analgesic activity in morphine are shown below





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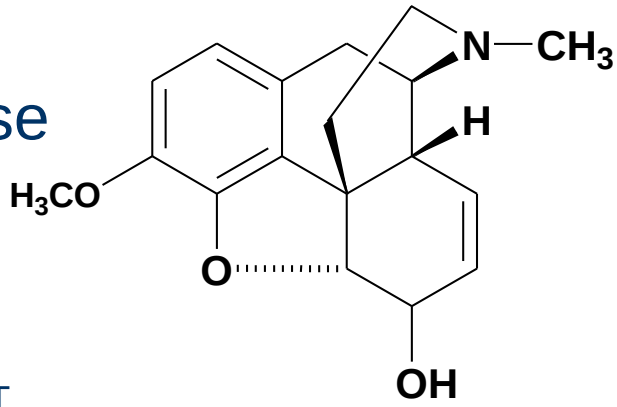
Classification of opoid analgesics

- Morphine and semi- synthetic derivatives
- Morphinan
- Benzomorphan
- 4- phenylpiperidine derivatives

Codeine

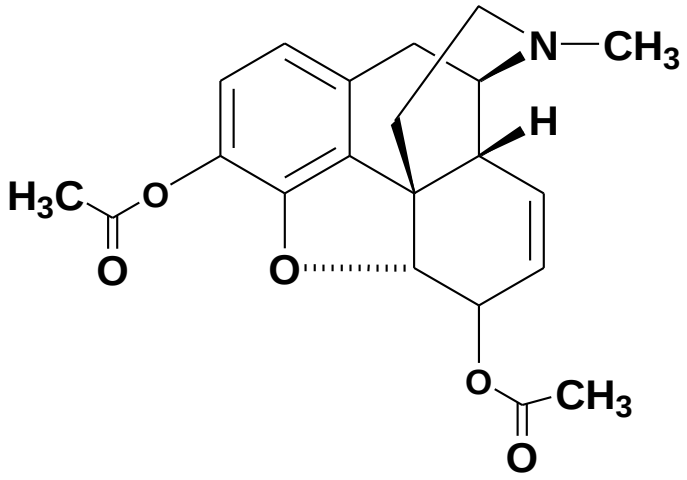
- Is a natural alkaloid
- Less active than morphine
  - Due to methylation of phenolic OH
- Demethylation in vivo to morphine to be active
- Is potent antitussive agent

  -Extensively used for this purpose



Heroin

- Is a highly addictive drug
  - More lipophilic than morphine
- Hydrolysis leads to generation of morphine

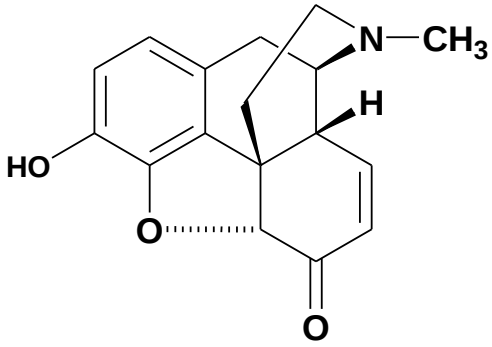


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- Introduced into market in 1898
- Withdrawn from general distribution
  - Due to addiction
- Now used in patients about to die of cancer
  - As analgesic
  - To counter depression faced by patient close to death

hydromorphinone

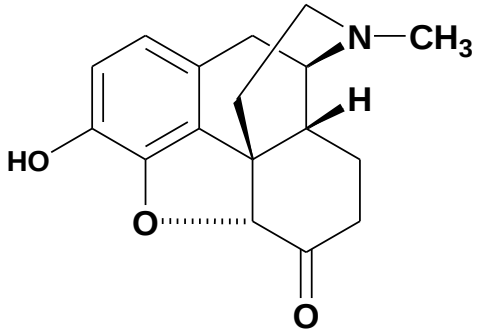
- Oxidation of C6 – OH to a ketone yields hydromorphinone with decreased side effects



Dihydromorphinone

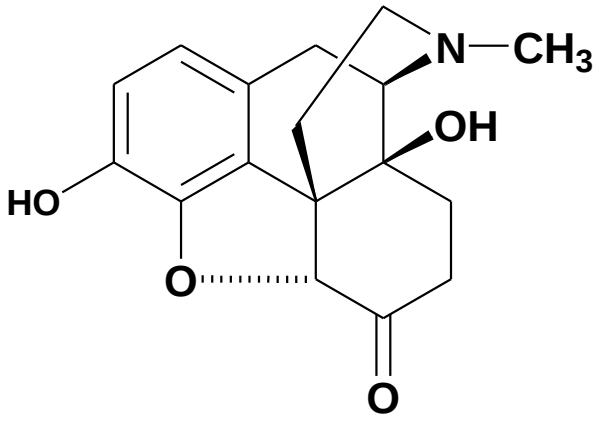
- formed by
  - Oxidation of C6 – OH to a ketone and
  - Saturation of C7 – C8 double bond
- Oxidation of the 6-OH of morphine

(without reduction of the 7, 8-double bond) does not significantly alter analgesic activity.



Dihydrohydroxymorphinone

- Hydroxyl on C-14 of dihydromorphinone
  - Used for the same purpose as morphine but with low antitussive activity

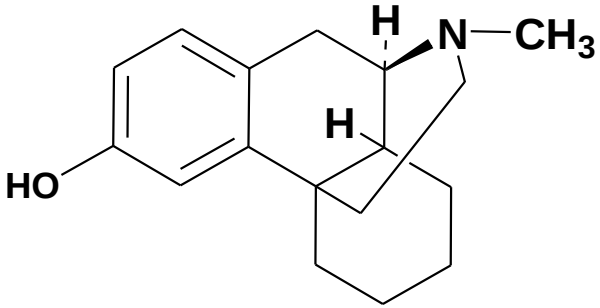


Activity
2-3x that of morphine

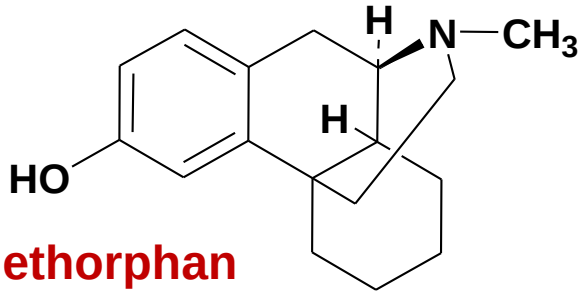
Morphinan

Activity:
5x that of morphine

- Analogues of morphine with loss of
  - 4,5- epoxide
  - C7 – C8 double bond
  - C-6 hydroxyl of morphine
- Levorphanol is a morphinan
  - (-) levorphanol is more potent than morphine
  - (+) levorphanol (dextrophan) lacks the proper configuration for bonding
- Serve as anti-tussive



Levorphanol

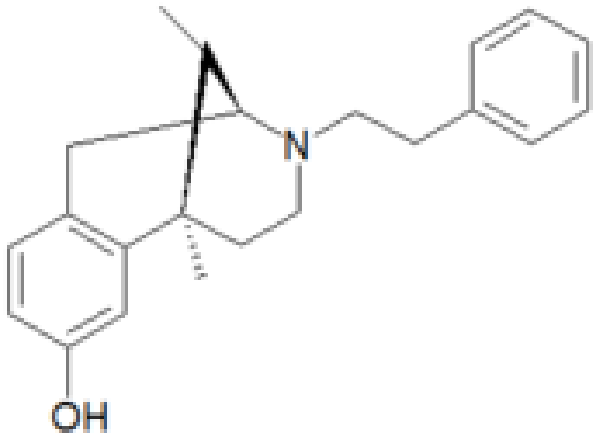


Dextromethorphan

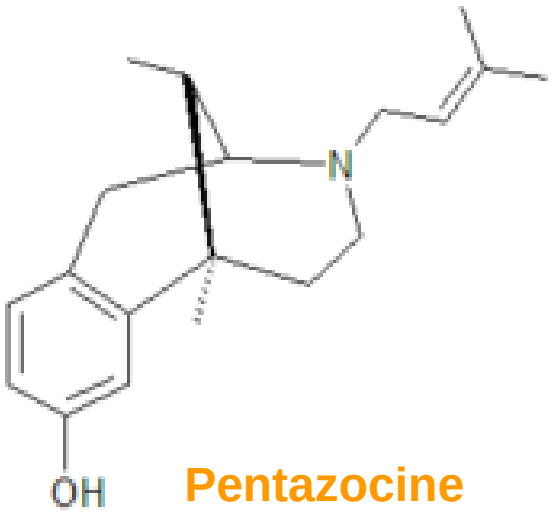
Benzomorphan

- Synthetic analogues lacking both
  - Epoxide ring
  - Ring C of morphine

Activity
4x that of morphine



Phenazocine



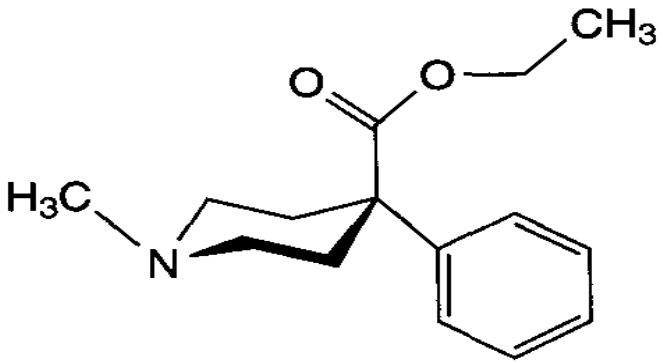
Pentazocine

Medchem,Alex T.

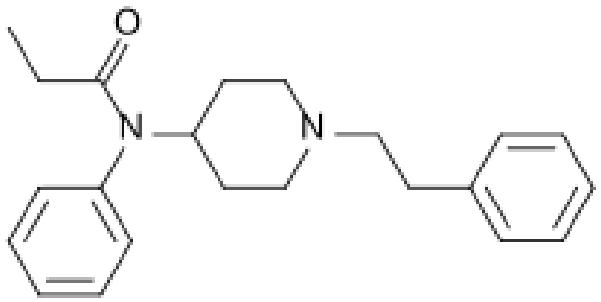
4- phenylpiperidine derivatives (Synthetic morphine derivatives)

- Simplification of the opoid pharmacophore led to development of phenylpiperidine derivatives

Activity:
1/10x that of morphine

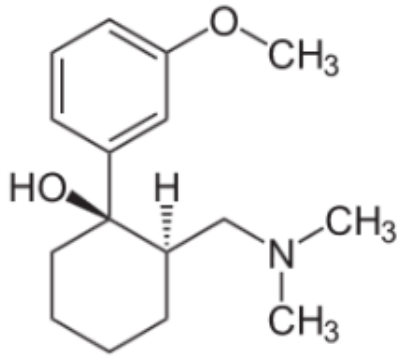


Meperidine (Pethidine)



Fentanyl

Medchem,Alex T.

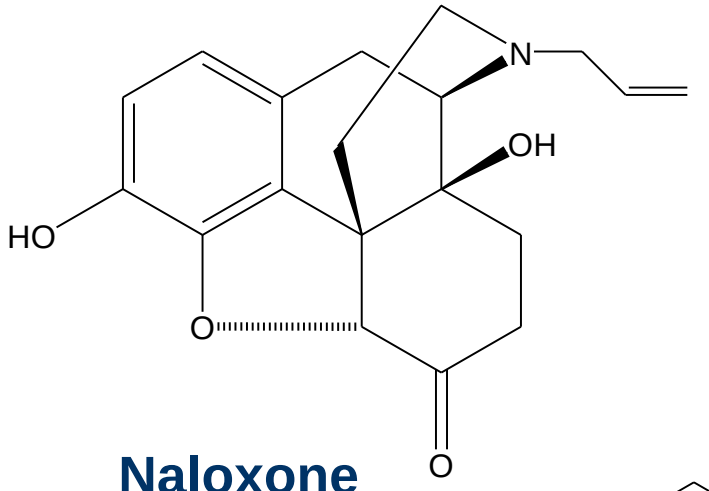


Tramadol

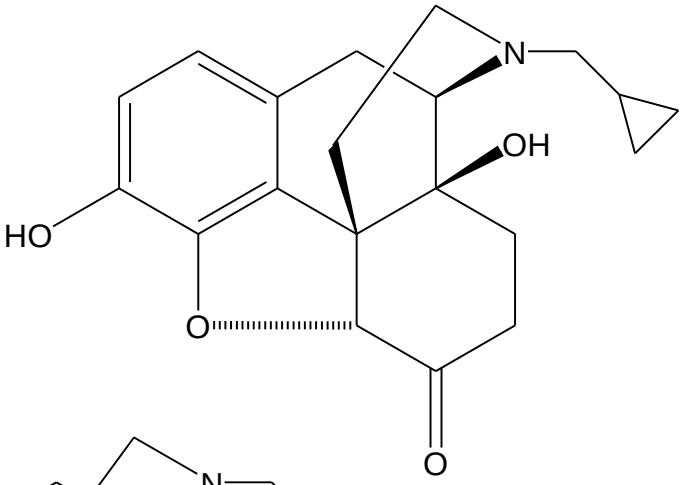
- Tramadol, a centrally acting analgesic, is used to treat moderate to severe pain
- Tramadol appears to act both as a serotonin releaser and as a serotonin reuptake inhibitor, and as a reuptake inhibitor of noradrenaline in vitro
- Tramadol is still an Opioid and thus highly addictive
- As an analgesic, tramadol is approximately equipotent as codeine

Opoid antagonists

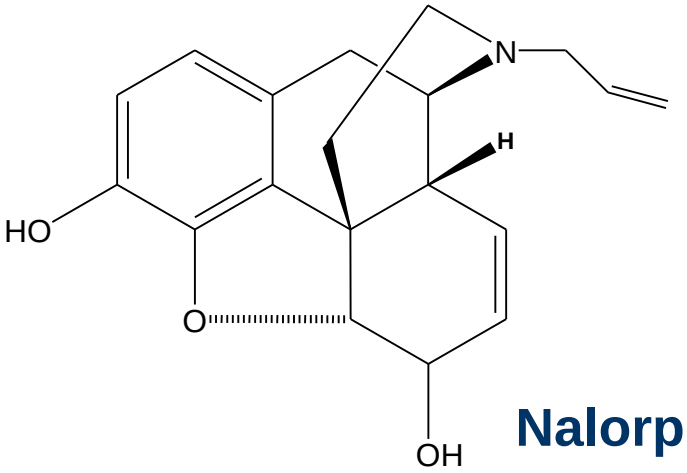
- Are obtained by replacing
  - N-methyl group with
    - N-allyl (-NCH_3 $CH= CH_2$) or N- cyclopropylmethyl
- Are used as antidote to reverse the adverse effects of agonists
  - Respiratory depression
  - Cardiovascular effects



Naloxone



Naltrexone



Nalorphine